

divides ab , that is, $\overline{ab} = \bar{0}$ in $\mathbb{Z}/n\mathbb{Z}$. This shows that every nonzero element of $\mathbb{Z}/n\mathbb{Z}$ is either a unit or a zero divisor. Furthermore, every nonzero element is a unit if and only if every integer a in the range $0 < a < n$ is relatively prime to n . This happens if and only if n is a prime, i.e., $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is a prime.

- (3) If R is the ring of all functions from the closed interval $[0, 1]$ to $\mathbb{R}$ then the units of R are the functions that are not zero at any point (for such f its inverse is the function $\frac{1}{f}$). If f is not a unit and not zero then f is a zero divisor because if we define

$$g(x) = \begin{cases} 0, & \text{if } f(x) \neq 0 \\ 1, & \text{if } f(x) = 0 \end{cases}$$

then g is not the zero function but $f(x)g(x) = 0$ for all x .

- (4) If R is the ring of all continuous functions from the closed interval $[0, 1]$ to $\mathbb{R}$ then the units of R are still the functions that are not zero at any point, but now there are functions that are neither units nor zero divisors. For instance, $f(x) = x - \frac{1}{2}$ has only one zero (at $x = \frac{1}{2}$) so f is not a unit. On the other hand, if $gf = 0$ then g must be zero for all $x \neq \frac{1}{2}$, and the only continuous function with this property is the zero function. Hence f is neither a unit nor a zero divisor. Similarly, no function with only a finite (or countable) number of zeros on $[0, 1]$ is a zero divisor. This ring also contains many zero divisors. For instance let

$$f(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2} \\ x - \frac{1}{2}, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

and let $g(x) = f(1 - x)$. Then f and g are nonzero continuous functions whose product is the zero function.

- (5) Let D be a rational number that is not a perfect square in $\mathbb{Q}$ and define

$$\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

as a subset of $\mathbb{C}$. This set is clearly closed under subtraction, and the identity $(a + b\sqrt{D})(c + d\sqrt{D}) = (ac + bdD) + (ad + bc)\sqrt{D}$ shows that it is also closed under multiplication. Hence $\mathbb{Q}(\sqrt{D})$ is a subring of $\mathbb{C}$ (even a subring of $\mathbb{R}$ if $D > 0$), so in particular is a commutative ring with identity. It is easy to show that the assumption that D is not a square implies that every element of $\mathbb{Q}(\sqrt{D})$ may be written uniquely in the form $a + b\sqrt{D}$. This assumption also implies that if a and b are not both 0 then $a^2 - Db^2$ is nonzero, and since $(a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2$ it follows that if $a + b\sqrt{D} \neq 0$ (i.e., one of a or b is nonzero) then $\frac{a - b\sqrt{D}}{a^2 - Db^2}$ is the inverse of $a + b\sqrt{D}$ in $\mathbb{Q}(\sqrt{D})$. This shows that every nonzero element in this commutative ring is a unit, i.e., $\mathbb{Q}(\sqrt{D})$ is a field (called a *quadratic field*, cf. Section 13.2).

The rational number D may be written $D = f^2 D'$ for some rational number f and a unique integer D' where D' is not divisible by the square of any integer greater than 1, i.e., D' is either -1 or ± 1 times the product of distinct primes in $\mathbb{Z}$ (for example, $8/5 = (2/5)^2 \cdot 10$). Call D' the *squarefree part* of D . Then $\sqrt{D} = f\sqrt{D'}$, and so $\mathbb{Q}(\sqrt{D}) = \mathbb{Q}(\sqrt{D'})$. Thus there is no loss in assuming that D is a squarefree integer (i.e., $f = 1$) in the definition of the quadratic field $\mathbb{Q}(\sqrt{D})$.

Rings having some of the same characteristics as the integers $\mathbb{Z}$ are given a name:

Definition. A commutative ring with identity $1 \neq 0$ is called an *integral domain* if it has no zero divisors.

The absence of zero divisors in integral domains give these rings a cancellation property:

Proposition 2. Assume a , b and c are elements of any ring with a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$ (i.e., if $a \neq 0$ we can cancel the a 's). In particular, if a , b , c are any elements in an integral domain and $ab = ac$, then either $a = 0$ or $b = c$.

Proof: If $ab = ac$ then $a(b - c) = 0$ so either $a = 0$ or $b - c = 0$. The second statement follows from the first and the definition of an integral domain.

Corollary 3. Any finite integral domain is a field.

Proof: Let R be a finite integral domain and let a be a nonzero element of R . By the cancellation law the map $x \mapsto ax$ is an injective function. Since R is finite this map is also surjective. In particular, there is some $b \in R$ such that $ab = 1$, i.e., a is a unit in R . Since a was an arbitrary nonzero element, R is a field.

A remarkable result of Wedderburn is that a finite division ring is necessarily commutative, i.e., is a field. A proof of this theorem is outlined in the exercises at the end of Section 13.6.

In Section 5 we study the relation between zero divisors and units in greater detail. We shall see that every nonzero element of a commutative ring that is not a zero divisor has a multiplicative inverse in some larger ring. This gives another perspective on the cancellation law in Proposition 2.

Having defined the notion of a ring, there is a natural notion of a subring.

Definition. A *subring* of the ring R is a subgroup of R that is closed under multiplication.

In other words, a subset S of a ring R is a subring if the operations of addition and multiplication in R when restricted to S give S the structure of a ring. To show that a subset of a ring R is a subring it suffices to check that it is *nonempty* and *closed under subtraction and under multiplication*.

Examples

A number of the examples above were also subrings.

- (1) $\mathbb{Z}$ is a subring of $\mathbb{Q}$ and $\mathbb{Q}$ is a subring of $\mathbb{R}$. The property “is a subring of” is clearly transitive.
- (2) $2\mathbb{Z}$ is a subring of $\mathbb{Z}$, as is $n\mathbb{Z}$ for any integer n . The ring $\mathbb{Z}/n\mathbb{Z}$ is not a subring (or a subgroup) of $\mathbb{Z}$ for any $n \geq 2$.

- (3) The ring of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ is a subring of the ring of all functions from $\mathbb{R}$ to $\mathbb{R}$. The ring of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ is a subring of both of these.
- (4) $S = \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}k$, the *integral* Quaternions, form a subring of either the real or the rational Quaternions — it is easy to check that multiplying two such quaternions together gives another quaternion with integer coefficients. This ring (which is not a division ring) can be used to give proofs for a number of results in number theory.
- (5) If R is a subring of a field F that contains the identity of F then R is an integral domain. The converse of this is also true, namely any integral domain is contained in a field (cf. Section 5).

Example: (Quadratic Integer Rings)

Let D be a squarefree integer. It is immediate from the addition and multiplication that the subset $\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$ forms a subring of the quadratic field $\mathbb{Q}(\sqrt{D})$ defined earlier. If $D \equiv 1 \pmod{4}$ then the slightly larger subset

$$\mathbb{Z}\left[\frac{1+\sqrt{D}}{2}\right] = \left\{a + b\frac{1+\sqrt{D}}{2} \mid a, b \in \mathbb{Z}\right\}$$

is also a subring: closure under addition is immediate and $(a + b\frac{1+\sqrt{D}}{2})(c + d\frac{1+\sqrt{D}}{2}) = (ac + bd\frac{D-1}{4}) + (ad + bc + bd)\frac{1+\sqrt{D}}{2}$ together with the congruence on D shows closure under multiplication.

Define

$$\mathcal{O} = \mathcal{O}_{\mathbb{Q}(\sqrt{D})} = \mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\},$$

where

$$\omega = \begin{cases} \sqrt{D}, & \text{if } D \equiv 2, 3 \pmod{4} \\ \frac{1+\sqrt{D}}{2}, & \text{if } D \equiv 1 \pmod{4}, \end{cases}$$

called the *ring of integers* in the quadratic field $\mathbb{Q}(\sqrt{D})$. The terminology comes from the fact that the elements of the subring $\mathcal{O}$ of the field $\mathbb{Q}(\sqrt{D})$ have many properties analogous to those of the subring of integers $\mathbb{Z}$ in the field of rational numbers $\mathbb{Q}$ (and are the *integral closure* of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{D})$ as explained in Section 15.3).

In the special case when $D = -1$ we obtain the ring $\mathbb{Z}[i]$ of *Gaussian integers*, which are the complex numbers $a + bi \in \mathbb{C}$ with a and b both *integers*. These numbers were originally introduced by Gauss around 1800 in order to state the biquadratic reciprocity law which deals with the beautiful relations that exist among fourth powers modulo primes. We shall shortly see another useful application of the algebraic structure of this ring to number theoretic questions.

Define the *field norm* $N : \mathbb{Q}(\sqrt{D}) \rightarrow \mathbb{Q}$ by

$$N(a + b\sqrt{D}) = (a + b\sqrt{D})(a - b\sqrt{D}) = a^2 - Db^2 \in \mathbb{Q},$$

which, as previously mentioned, is nonzero if $a + b\sqrt{D} \neq 0$. This norm gives a measure of “size” in the field $\mathbb{Q}(\sqrt{D})$. For instance when $D = -1$ the norm of $a + bi$ is $a^2 + b^2$, which is the square of the length of this complex number considered as a vector in the complex plane. We shall use the field norm in this and subsequent examples to establish many properties of the rings $\mathcal{O}$.

It is easy to check that N is *multiplicative*, i.e., that $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in \mathbb{Q}(\sqrt{D})$. On the subring $\mathcal{O}$ it is also easy to see that the field norm is given by

$$N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = \begin{cases} a^2 - Db^2, & \text{if } D \equiv 2, 3 \pmod{4} \\ a^2 + ab + \frac{1-D}{4}b^2, & \text{if } D \equiv 1 \pmod{4} \end{cases}$$

where

$$\bar{\omega} = \begin{cases} -\sqrt{D}, & \text{if } D \equiv 2, 3 \pmod{4} \\ \frac{1-\sqrt{D}}{2}, & \text{if } D \equiv 1 \pmod{4}. \end{cases}$$

It follows that $N(\alpha)$ is in fact an *integer* for every $\alpha \in \mathcal{O}$.

We may use this norm to characterize the units in $\mathcal{O}$. If $\alpha \in \mathcal{O}$ has field norm $N(\alpha) = \pm 1$, the previous formula shows that $(a + b\omega)^{-1} = \pm(a + b\bar{\omega})$, which is again an element of $\mathcal{O}$ and so α is a unit in $\mathcal{O}$. Suppose conversely that α is a unit in $\mathcal{O}$, say $\alpha\beta = 1$ for some $\beta \in \mathcal{O}$. Then the multiplicative property of the field norm implies that $N(\alpha)N(\beta) = N(\alpha\beta) = N(1) = 1$. Since both $N(\alpha)$ and $N(\beta)$ are integers, each must be ± 1 . Hence,

the element α is a unit in $\mathcal{O}$ if and only if $N(\alpha) = \pm 1$.

In particular the determination of the integer solutions to the equation $x^2 - Dy^2 = \pm 1$ (called *Pell's equation* in elementary number theory) is essentially equivalent to the determination of the units in the ring $\mathcal{O}$.

When $D = -1$, the units in the Gaussian integers $\mathbb{Z}[i]$ are the elements $a + bi$ with $a^2 + b^2 = \pm 1$, $a, b \in \mathbb{Z}$, so the group of units consists of $\{\pm 1, \pm i\}$. When $D = -3$, the units in $\mathbb{Z}[(1 + \sqrt{-3})/2]$ are determined by the integers a, b with $a^2 + ab + b^2 = \pm 1$, i.e., with $(2a + b)^2 + 3b^2 = \pm 4$, from which it is easy to see that the group of units is a group of order 6 given by $\{\pm 1, \pm \rho, \pm \rho^2\}$ where $\rho = (-1 + \sqrt{-3})/2$. For any other $D < 0$ it is similarly straightforward to see that the only units are $\{\pm 1\}$.

By contrast, when $D > 0$ it can be shown that the group of units $\mathcal{O}^\times$ is always infinite. For example, it is easy to check that $1 + \sqrt{2}$ is a unit in the ring $\mathcal{O} = \mathbb{Z}[\sqrt{2}]$ (with field norm -1) and that $\{\pm(1 + \sqrt{2})^n \mid n \in \mathbb{Z}\}$ is an infinite set of distinct units (in fact the full group of units in this case, but this is harder to prove).

EXERCISES

Let R be a ring with 1.

1. Show that $(-1)^2 = 1$ in R .
2. Prove that if u is a unit in R then so is $-u$.
3. Let R be a ring with identity and let S be a subring of R containing the identity. Prove that if u is a unit in S then u is a unit in R . Show by example that the converse is false.
4. Prove that the intersection of any nonempty collection of subrings of a ring is also a subring.
5. Decide which of the following (a) – (f) are subrings of $\mathbb{Q}$:
 - (a) the set of all rational numbers with odd denominators (when written in lowest terms)
 - (b) the set of all rational numbers with even denominators (when written in lowest terms)
 - (c) the set of nonnegative rational numbers
 - (d) the set of squares of rational numbers
 - (e) the set of all rational numbers with odd numerators (when written in lowest terms)

- (f) the set of all rational numbers with even numerators (when written in lowest terms).
6. Decide which of the following are subrings of the ring of all functions from the closed interval $[0,1]$ to $\mathbb{R}$:
- (a) the set of all functions $f(x)$ such that $f(q) = 0$ for all $q \in \mathbb{Q} \cap [0, 1]$
 - (b) the set of all polynomial functions
 - (c) the set of all functions which have only a finite number of zeros, together with the zero function
 - (d) the set of all functions which have an infinite number of zeros
 - (e) the set of all functions f such that $\lim_{x \rightarrow 1^-} f(x) = 0$
 - (f) the set of all rational linear combinations of the functions $\sin nx$ and $\cos mx$, where $m, n \in \{0, 1, 2, \dots\}$.
7. The *center* of a ring R is $\{z \in R \mid zr = rz \text{ for all } r \in R\}$ (i.e., is the set of all elements which commute with every element of R). Prove that the center of a ring is a subring that contains the identity. Prove that the center of a division ring is a field.
8. Describe the center of the real Hamilton Quaternions $\mathbb{H}$. Prove that $\{a + bi \mid a, b \in \mathbb{R}\}$ is a subring of $\mathbb{H}$ which is a field but is not contained in the center of $\mathbb{H}$.
9. For a fixed element $a \in R$ define $C(a) = \{r \in R \mid ra = ar\}$. Prove that $C(a)$ is a subring of R containing a . Prove that the center of R is the intersection of the subrings $C(a)$ over all $a \in R$.
10. Prove that if D is a division ring then $C(a)$ is a division ring for all $a \in D$ (cf. the preceding exercise).
11. Prove that if R is an integral domain and $x^2 = 1$ for some $x \in R$ then $x = \pm 1$.
12. Prove that any subring of a field which contains the identity is an integral domain.
13. An element x in R is called *nilpotent* if $x^m = 0$ for some $m \in \mathbb{Z}^+$.
- (a) Show that if $n = a^k b$ for some integers a and b then $\bar{a}b$ is a nilpotent element of $\mathbb{Z}/n\mathbb{Z}$.
 - (b) If $a \in \mathbb{Z}$ is an integer, show that the element $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is nilpotent if and only if every prime divisor of n is also a divisor of a . In particular, determine the nilpotent elements of $\mathbb{Z}/72\mathbb{Z}$ explicitly.
 - (c) Let R be the ring of functions from a nonempty set X to a field F . Prove that R contains no nonzero nilpotent elements.
14. Let x be a nilpotent element of the commutative ring R (cf. the preceding exercise).
- (a) Prove that x is either zero or a zero divisor.
 - (b) Prove that rx is nilpotent for all $r \in R$.
 - (c) Prove that $1 + x$ is a unit in R .
 - (d) Deduce that the sum of a nilpotent element and a unit is a unit.
15. A ring R is called a *Boolean ring* if $a^2 = a$ for all $a \in R$. Prove that every Boolean ring is commutative.
16. Prove that the only Boolean ring that is an integral domain is $\mathbb{Z}/2\mathbb{Z}$.
17. Let R and S be rings. Prove that the direct product $R \times S$ is a ring under componentwise addition and multiplication. Prove that $R \times S$ is commutative if and only if both R and S are commutative. Prove that $R \times S$ has an identity if and only if both R and S have identities.
18. Prove that $\{(r, r) \mid r \in R\}$ is a subring of $R \times R$.
19. Let I be any nonempty index set and let R_i be a ring for each $i \in I$. Prove that the direct

product $\prod_{i \in I} R_i$ is a ring under componentwise addition and multiplication.

20. Let R be the collection of sequences $(a_1, a_2, a_3, \dots)$ of integers $a_1, a_2, a_3, \dots$ where all but finitely many of the a_i are 0 (called the *direct sum* of infinitely many copies of $\mathbb{Z}$). Prove that R is a ring under componentwise addition and multiplication which does not have an identity.
21. Let X be any nonempty set and let $\mathcal{P}(X)$ be the set of all subsets of X (the *power set* of X). Define addition and multiplication on $\mathcal{P}(X)$ by

$$A + B = (A - B) \cup (B - A) \quad \text{and} \quad A \times B = A \cap B$$

i.e., addition is symmetric difference and multiplication is intersection.

- (a) Prove that $\mathcal{P}(X)$ is a ring under these operations ($\mathcal{P}(X)$ and its subrings are often referred to as *rings of sets*).
- (b) Prove that this ring is commutative, has an identity and is a Boolean ring.
22. Give an example of an infinite Boolean ring.
23. Let D be a squarefree integer, and let $\mathcal{O}$ be the ring of integers in the quadratic field $\mathbb{Q}(\sqrt{D})$. For any positive integer f prove that the set $\mathcal{O}_f = \mathbb{Z}[f\omega] = \{a + bf\omega \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathcal{O}$ containing the identity. Prove that $[\mathcal{O} : \mathcal{O}_f] = f$ (index as additive abelian groups). Prove conversely that a subring of $\mathcal{O}$ containing the identity and having finite index f in $\mathcal{O}$ (as additive abelian group) is equal to $\mathcal{O}_f$. (The ring $\mathcal{O}_f$ is called the *order of conductor f* in the field $\mathbb{Q}(\sqrt{D})$. The ring of integers $\mathcal{O}$ is called the *maximal order* in $\mathbb{Q}(\sqrt{D})$.)
24. Show for $D = 3, 5, 6$, and 7 that the group of units $\mathcal{O}^\times$ of the quadratic integer ring $\mathcal{O}$ is infinite by exhibiting an explicit unit of infinite (multiplicative) order in each ring.
25. Let I be the ring of integral Hamilton Quaternions and define

$$N : I \rightarrow \mathbb{Z} \quad \text{by} \quad N(a + bi + cj + dk) = a^2 + b^2 + c^2 + d^2$$

(the map N is called a *norm*).

- (a) Prove that $N(\alpha) = \alpha\bar{\alpha}$ for all $\alpha \in I$, where if $\alpha = a + bi + cj + dk$ then $\bar{\alpha} = a - bi - cj - dk$.
- (b) Prove that $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in I$.
- (c) Prove that an element of I is a unit if and only if it has norm $+1$. Show that $I^\times$ is isomorphic to the quaternion group of order 8. [The inverse in the ring of rational quaternions of a nonzero element α is $\frac{\bar{\alpha}}{N(\alpha)}$.]
26. Let K be a field. A *discrete valuation* on K is a function $v : K^\times \rightarrow \mathbb{Z}$ satisfying
- (i) $v(ab) = v(a) + v(b)$ (i.e., v is a homomorphism from the multiplicative group of nonzero elements of K to $\mathbb{Z}$),
 - (ii) v is surjective, and
 - (iii) $v(x + y) \geq \min\{v(x), v(y)\}$ for all $x, y \in K^\times$ with $x + y \neq 0$.
- The set $R = \{x \in K^\times \mid v(x) \geq 0\} \cup \{0\}$ is called the *valuation ring* of v .
- (a) Prove that R is a subring of K which contains the identity. (In general, a ring R is called a *discrete valuation ring* if there is some field K and some discrete valuation v on K such that R is the valuation ring of v .)
- (b) Prove that for each nonzero element $x \in K$ either x or x^{-1} is in R .
- (c) Prove that an element x is a unit of R if and only if $v(x) = 0$.
27. A specific example of a discrete valuation ring (cf. the preceding exercise) is obtained

when p is a prime, $K = \mathbb{Q}$ and

$$v_p : \mathbb{Q}^\times \rightarrow \mathbb{Z} \quad \text{by} \quad v_p\left(\frac{a}{b}\right) = \alpha \quad \text{where} \quad \frac{a}{b} = p^\alpha \frac{c}{d}, \quad p \nmid c \text{ and } p \nmid d.$$

Prove that the corresponding valuation ring R is the ring of all rational numbers whose denominators are relatively prime to p . Describe the units of this valuation ring.

- 28.** Let R be a ring with $1 \neq 0$. A nonzero element a is called a *left zero divisor* in R if there is a nonzero element $x \in R$ such that $ax = 0$. Symmetrically, $b \neq 0$ is a *right zero divisor* if there is a nonzero $y \in R$ such that $yb = 0$ (so a zero divisor is an element which is either a left or a right zero divisor). An element $u \in R$ has a *left inverse* in R if there is some $s \in R$ such that $su = 1$. Symmetrically, v has a *right inverse* if $vt = 1$ for some $t \in R$.
- (a) Prove that u is a unit if and only if it has both a right and a left inverse (i.e., u must have a two-sided inverse).
 - (b) Prove that if u has a right inverse then u is not a right zero divisor.
 - (c) Prove that if u has more than one right inverse then u is a left zero divisor.
 - (d) Prove that if R is a finite ring then every element that has a right inverse is a unit (i.e., has a two-sided inverse).
- 29.** Let A be any commutative ring with identity $1 \neq 0$. Let R be the set of all group homomorphisms of the additive group A to itself with addition defined as pointwise addition of functions and multiplication defined as function composition. Prove that these operations make R into a ring with identity. Prove that the units of R are the group automorphisms of A (cf. Exercise 20, Section 1.6).
- 30.** Let $A = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \cdots$ be the direct product of copies of $\mathbb{Z}$ indexed by the positive integers (so A is a ring under componentwise addition and multiplication) and let R be the ring of all group homomorphisms from A to itself as described in the preceding exercise. Let φ be the element of R defined by $\varphi(a_1, a_2, a_3, \dots) = (a_2, a_3, \dots)$. Let ψ be the element of R defined by $\psi(a_1, a_2, a_3, \dots) = (0, a_1, a_2, a_3, \dots)$.
- (a) Prove that $\varphi\psi$ is the identity of R but $\psi\varphi$ is not the identity of R (i.e., ψ is a *right inverse* for φ but not a left inverse).
 - (b) Exhibit infinitely many right inverses for φ .
 - (c) Find a nonzero element π in R such that $\varphi\pi = 0$ but $\pi\varphi \neq 0$.
 - (d) Prove that there is no nonzero element $\lambda \in R$ such that $\lambda\varphi = 0$ (i.e., φ is a left zero divisor but not a right zero divisor).

7.2 EXAMPLES: POLYNOMIAL RINGS, MATRIX RINGS, AND GROUP RINGS

We introduce here three important types of rings: polynomial rings, matrix rings, and group rings. We shall see in the course of the text that these three classes of rings are often related. For example, we shall see in Part VI that the group ring of a group G over the complex numbers $\mathbb{C}$ is a direct product of matrix rings over $\mathbb{C}$.

These rings also have many important applications, in addition to being interesting in their own right. In Part III we shall use polynomial rings to prove some classification theorems for matrices which, in particular, determine when a matrix is similar to a diagonal matrix. In Part VI we shall use group rings to study group actions and to prove some additional important classification theorems.

Polynomial Rings

Fix a commutative ring R with identity. We define the ring of polynomials in a form which may already be familiar, at least for polynomials with real coefficients. A definition in terms of Cartesian products is given in Appendix I. Let x be an indeterminate. The formal sum

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

with $n \geq 0$ and each $a_i \in R$ is called a *polynomial* in x with coefficients a_i in R . If $a_n \neq 0$, then the polynomial is said to be of *degree* n , $a_n x^n$ is called the *leading term*, and a_n is called the *leading coefficient* (where the leading coefficient of the zero polynomial is taken to be 0). The polynomial is *monic* if $a_n = 1$. The set of all such polynomials is called the ring of *polynomials in the variable x with coefficients in R* and will be denoted $R[x]$.

The operations of addition and multiplication which make $R[x]$ into a ring are the same operations familiar from elementary algebra: addition is “componentwise”

$$\begin{aligned} (a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0) &+ (b_n x^n + b_{n-1} x^{n-1} + \cdots + b_1 x + b_0) \\ &= (a_n + b_n) x^n + (a_{n-1} + b_{n-1}) x^{n-1} + \cdots + (a_1 + b_1) x + (a_0 + b_0) \end{aligned}$$

(here a_n or b_n may be zero in order for addition of polynomials of different degrees to be defined). Multiplication is performed by first defining $(ax^i)(bx^j) = abx^{i+j}$ for polynomials with only one nonzero term and then extending to all polynomials by the distributive laws (usually referred to as “expanding out and collecting like terms”):

$$\begin{aligned} (a_0 + a_1 x + a_2 x^2 + \cdots) &\times (b_0 + b_1 x + b_2 x^2 + \cdots) \\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0) x + (a_0 b_2 + a_1 b_1 + a_2 b_0) x^2 + \cdots \end{aligned}$$

(in general, the coefficient of x^k in the product will be $\sum_{i=0}^k a_i b_{k-i}$). These operations make sense since R is a ring so the sums and products of the coefficients are defined. An easy verification proves that $R[x]$ is indeed a ring with these definitions of addition and multiplication.

The ring R appears in $R[x]$ as the *constant polynomials*. Note that by definition of the multiplication, $R[x]$ is a *commutative ring with identity* (the identity 1 from R).

The coefficient ring R above was assumed to be a commutative ring since that is the situation we shall be primarily interested in, but note that the definition of the addition and multiplication in $R[x]$ above would be valid even if R were not commutative or did not have an identity. If the coefficient ring R is the integers $\mathbb{Z}$ (respectively, the rationals $\mathbb{Q}$) the polynomial ring $\mathbb{Z}[x]$ (respectively, $\mathbb{Q}[x]$) is the ring of polynomials with integer (rational) coefficients familiar from elementary algebra.

Another example is the polynomial ring $\mathbb{Z}/3\mathbb{Z}[x]$ of polynomials in x with coefficients in $\mathbb{Z}/3\mathbb{Z}$. This ring consists of nonnegative powers of x with coefficients 0, 1, and 2 with calculations on the coefficients performed modulo 3. For example, if

$$p(x) = x^2 + 2x + 1 \quad \text{and} \quad q(x) = x^3 + x + 2$$

then

$$p(x) + q(x) = x^3 + x^2$$

and

$$p(x)q(x) = x^5 + 2x^4 + 2x^3 + x^2 + 2x + 2.$$

The ring in which the coefficients are taken makes a substantial difference in the behavior of polynomials. For example, the polynomial $x^2 + 1$ is not a perfect square in the polynomial ring $\mathbb{Z}[x]$, but is a perfect square in the polynomial ring $\mathbb{Z}/2\mathbb{Z}[x]$, since $(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1$ in this ring.

Proposition 4. Let R be an integral domain and let $p(x), q(x)$ be nonzero elements of $R[x]$. Then

- (1) $\text{degree } p(x)q(x) = \text{degree } p(x) + \text{degree } q(x)$,
- (2) the units of $R[x]$ are just the units of R ,
- (3) $R[x]$ is an integral domain.

Proof: If R has no zero divisors then neither does $R[x]$; if $p(x)$ and $q(x)$ are polynomials with leading terms $a_n x^n$ and $b_m x^m$, respectively, then the leading term of $p(x)q(x)$ is $a_n b_m x^{n+m}$, and $a_n b_m \neq 0$. This proves (3) and also verifies (1). If $p(x)$ is a unit, say $p(x)q(x) = 1$ in $R[x]$, then $\text{degree } p(x) + \text{degree } q(x) = 0$, so both $p(x)$ and $q(x)$ are elements of R , hence are units in R since their product is 1. This proves (2).

If the ring R has zero divisors then so does $R[x]$, because $R \subset R[x]$. Also, if $f(x)$ is a zero divisor in $R[x]$ (i.e., $f(x)g(x) = 0$ for some nonzero $g(x) \in R[x]$) then in fact $cf(x) = 0$ for some nonzero $c \in R$ (cf. Exercise 2).

If S is a subring of R then $S[x]$ is a subring of $R[x]$. For instance, $\mathbb{Z}[x]$ is a subring of $\mathbb{Q}[x]$. Some other examples of subrings of $R[x]$ are the set of all polynomials in x^2 (i.e., in which only even powers of x appear) and the set of all polynomials with zero constant term (the latter subring does not have an identity).

Polynomial rings, particularly those over fields, will be studied extensively in Chapter 9.

Matrix Rings

Fix an arbitrary ring R and let n be a positive integer. Let $M_n(R)$ be the set of all $n \times n$ matrices with entries from R . The element (a_{ij}) of $M_n(R)$ is an $n \times n$ square array of elements of R whose entry in row i and column j is $a_{ij} \in R$. The set of matrices becomes a ring under the usual rules by which matrices of real numbers are added and multiplied. Addition is componentwise: the i, j entry of the matrix $(a_{ij}) + (b_{ij})$ is $a_{ij} + b_{ij}$. The i, j entry of the matrix product $(a_{ij}) \times (b_{ij})$ is $\sum_{k=1}^n a_{ik} b_{kj}$ (note that these matrices need to be square in order that multiplication of any two elements be defined). It is a straightforward calculation to check that these operations make $M_n(R)$ into a ring. When R is a field we shall prove that $M_n(R)$ is a ring by less computational means in Part III.

Note that if R is any nontrivial ring (even a commutative one) and $n \geq 2$ then $M_n(R)$ is *not commutative*: if $ab \neq 0$ in R let A be the matrix with a in position 1,1 and zeros elsewhere and let B be the matrix with b in position 1,2 and zeros elsewhere; then AB is the (nonzero) entry in position 1,2 of AB whereas BA is the zero matrix.

These two matrices also show that $M_n(R)$ has zero divisors for all nonzero rings R whenever $n \geq 2$.

An element (a_{ij}) of $M_n(R)$ is called a *scalar matrix* if for some $a \in R$, $a_{ii} = a$ for all $i \in \{1, \dots, n\}$ and $a_{ij} = 0$ for all $i \neq j$ (i.e., all diagonal entries equal a and all off-diagonal entries are 0). The set of scalar matrices is a subring of $M_n(R)$. This subring is a copy of R (i.e., is “isomorphic” to R): if the matrix A has the element a along the main diagonal and the matrix B has the element b along the main diagonal then the matrix $A + B$ has $a + b$ along the diagonal and AB has ab along the diagonal (and all other entries 0). If R is commutative, the scalar matrices commute with all elements of $M_n(R)$. If R has a 1, then the scalar matrix with 1’s down the diagonal (the $n \times n$ identity matrix) is the 1 of $M_n(R)$. In this case the units in $M_n(R)$ are the invertible $n \times n$ matrices and the group of units is denoted $GL_n(R)$, the *general linear group* of degree n over R .

If S is a subring of R then $M_n(S)$ is a subring of $M_n(R)$. For instance $M_n(\mathbb{Z})$ is a subring of $M_n(\mathbb{Q})$ and $M_n(2\mathbb{Z})$ is a subring of both of these. Another example of a subring of $M_n(R)$ is the set of *upper triangular* matrices: $\{(a_{ij}) \mid a_{pq} = 0 \text{ whenever } p > q\}$ (the set of matrices all of whose entries below the main diagonal are zero) — one easily checks that the sum and product of upper triangular matrices is upper triangular.

Group Rings

Fix a commutative ring R with identity $1 \neq 0$ and let $G = \{g_1, g_2, \dots, g_n\}$ be any finite group with group operation written multiplicatively. Define the *group ring*, RG , of G with coefficients in R to be the set of all formal sums

$$a_1g_1 + a_2g_2 + \cdots + a_ng_n \quad a_i \in R, \quad 1 \leq i \leq n.$$

If g_1 is the identity of G we shall write a_1g_1 simply as a_1 . Similarly, we shall write the element $1g$ for $g \in G$ simply as g .

Addition is defined “componentwise”

$$\begin{aligned} (a_1g_1 + a_2g_2 + \cdots + a_ng_n) + (b_1g_1 + b_2g_2 + \cdots + b_ng_n) \\ = (a_1 + b_1)g_1 + (a_2 + b_2)g_2 + \cdots + (a_n + b_n)g_n. \end{aligned}$$

Multiplication is performed by first defining $(ag_i)(bg_j) = (ab)g_k$, where the product ab is taken in R and $g_i g_j = g_k$ is the product in the group G . This product is then extended to all formal sums by the distributive laws so that the coefficient of g_k in the product $(a_1g_1 + \cdots + a_ng_n) \times (b_1g_1 + \cdots + b_ng_n)$ is $\sum_{g_i g_j = g_k} a_i b_j$. It is straightforward to check that these operations make RG into a ring (again, commutativity of R is not needed). The associativity of multiplication follows from the associativity of the group operation in G . The ring RG is commutative if and only if G is a commutative group.

Example

Let $G = D_8$ be the dihedral group of order 8 with the usual generators r, s ($r^4 = s^2 = 1$ and $rs = sr^{-1}$) and let $R = \mathbb{Z}$. The elements $\alpha = r + r^2 - 2s$ and $\beta = -3r^2 + rs$ are

typical members of $\mathbb{Z}D_8$. Their sum and product are then

$$\begin{aligned}\alpha + \beta &= r - 2r^2 - 2s + rs \\ \alpha\beta &= (r + r^2 - 2s)(-3r^2 + rs) \\ &= r(-3r^2 + rs) + r^2(-3r^2 + rs) - 2s(-3r^2 + rs) \\ &= -3r^3 + r^2s - 3 + r^3s + 6r^2s - 2r^3 \\ &= -3 - 5r^3 + 7r^2s + r^3s.\end{aligned}$$

The ring R appears in RG as the “constant” formal sums i.e., the R -multiples of the identity of G (note that the definition of the addition and multiplication in RG restricted to these elements is just the addition and multiplication in R). These elements of R commute with all elements of RG . The identity of R is the identity of RG .

The group G also appears in RG (the element g_i appears as $1g_i$ — for example, $r, s \in D_8$ are also elements of the group ring $\mathbb{Z}D_8$ above) — multiplication in the ring RG restricted to G is just the group operation. In particular, each element of G has a multiplicative inverse in the ring RG (namely, its inverse in G). This says that G is a *subgroup of the group of units of RG* .

If $|G| > 1$ then RG always has zero divisors. For example, let g be any element of G of order $m > 1$. Then

$$(1 - g)(1 + g + \cdots + g^{m-1}) = 1 - g^m = 1 - 1 = 0$$

so $1 - g$ is a zero divisor (note that by definition of RG neither of the formal sums in the above product is zero).

If S is a subring of R then SG is a subring of RG . For instance, $\mathbb{Z}G$ (called the *integral group ring of G*) is a subring of $\mathbb{Q}G$ (the *rational group ring of G*). Furthermore, if H is a subgroup of G then RH is a subring of RG . The set of all elements of RG whose coefficients sum to zero is a subring (without identity). If $|G| > 1$, the set of elements with zero “constant term” (i.e., the coefficient of the identity of G is zero) is *not* a subring (it is not closed under multiplication).

Note that the group ring $\mathbb{R}Q_8$ is *not* the same ring as the Hamilton Quaternions $\mathbb{H}$ even though the latter contains a copy of the quaternion group Q_8 (under multiplication). One difference is that the unique element of order 2 in Q_8 (usually denoted by -1) is not the additive inverse of 1 in $\mathbb{R}Q_8$. In other words, if we temporarily denote the identity of the group Q_8 by g_1 and the unique element of order 2 by g_2 , then $g_1 + g_2$ is not zero in $\mathbb{R}Q_8$, whereas $1 + (-1)$ is zero in $\mathbb{H}$. Furthermore, as noted above, the group ring $\mathbb{R}Q_8$ contains zero divisors hence is not a division ring.

Group rings over fields will be studied extensively in Chapter 18.

EXERCISES

Let R be a commutative ring with 1.

- Let $p(x) = 2x^3 - 3x^2 + 4x - 5$ and let $q(x) = 7x^3 + 33x - 4$. In each of parts (a), (b) and (c) compute $p(x) + q(x)$ and $p(x)q(x)$ under the assumption that the coefficients of the two given polynomials are taken from the specified ring (where the integer coefficients are taken mod n in parts (b) and (c)):
 (a) $R = \mathbb{Z}$, (b) $R = \mathbb{Z}/2\mathbb{Z}$, (c) $R = \mathbb{Z}/3\mathbb{Z}$.

2. Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$. Prove that $p(x)$ is a zero divisor in $R[x]$ if and only if there is a nonzero $b \in R$ such that $bp(x) = 0$. [Let $g(x) = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_0$ be a nonzero polynomial of minimal degree such that $g(x)p(x) = 0$. Show that $b_m a_n = 0$ and so $a_n g(x)$ is a polynomial of degree less than m that also gives 0 when multiplied by $p(x)$. Conclude that $a_n g(x) = 0$. Apply a similar argument to show by induction on i that $a_{n-i} g(x) = 0$ for $i = 0, 1, \dots, n$, and show that this implies $b_m p(x) = 0$.]
3. Define the set $R[[x]]$ of *formal power series* in the indeterminate x with coefficients from R to be all formal infinite sums

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

Define addition and multiplication of power series in the same way as for power series with real or complex coefficients i.e., extend polynomial addition and multiplication to power series as though they were “polynomials of infinite degree”:

$$\begin{aligned} \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} b_n x^n &= \sum_{n=0}^{\infty} (a_n + b_n) x^n \\ \sum_{n=0}^{\infty} a_n x^n \times \sum_{n=0}^{\infty} b_n x^n &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n. \end{aligned}$$

(The term “formal” is used here to indicate that convergence is not considered, so that formal power series need not represent functions on R .)

- (a) Prove that $R[[x]]$ is a commutative ring with 1.
 (b) Show that $1 - x$ is a unit in $R[[x]]$ with inverse $1 + x + x^2 + \cdots$.
 (c) Prove that $\sum_{n=0}^{\infty} a_n x^n$ is a unit in $R[[x]]$ if and only if a_0 is a unit in R .
4. Prove that if R is an integral domain then the ring of formal power series $R[[x]]$ is also an integral domain.
5. Let F be a field and define the ring $F((x))$ of *formal Laurent series* with coefficients from F by

$$F((x)) = \left\{ \sum_{n \geq N} a_n x^n \mid a_n \in F \text{ and } N \in \mathbb{Z} \right\}.$$

(Every element of $F((x))$ is a power series in x plus a polynomial in $1/x$, i.e., each element of $F((x))$ has only a finite number of terms with negative powers of x .)

- (a) Prove that $F((x))$ is a field.
 (b) Define the map

$$v : F((x))^{\times} \rightarrow \mathbb{Z} \quad \text{by} \quad v\left(\sum_{n \geq N} a_n x^n\right) = N$$

where a_N is the first nonzero coefficient of the series (i.e., N is the “order of zero or pole of the series at 0”). Prove that v is a discrete valuation on $F((x))$ whose discrete valuation ring is $F[[x]]$, the ring of formal power series (cf. Exercise 26, Section 1).

6. Let S be a ring with identity $1 \neq 0$. Let $n \in \mathbb{Z}^+$ and let A be an $n \times n$ matrix with entries from S whose i, j entry is a_{ij} . Let E_{ij} be the element of $M_n(S)$ whose i, j entry is 1 and whose other entries are all 0.

- (a) Prove that $E_{ij}A$ is the matrix whose i^{th} row equals the j^{th} row of A and all other rows are zero.
- (b) Prove that AE_{ij} is the matrix whose j^{th} column equals the i^{th} column of A and all other columns are zero.
- (c) Deduce that $E_{pq}AE_{rs}$ is the matrix whose p, s entry is a_{qr} and all other entries are zero.
7. Prove that the center of the ring $M_n(R)$ is the set of scalar matrices (cf. Exercise 7, Section 1). [Use the preceding exercise.]
8. Let S be any ring and let $n \geq 2$ be an integer. Prove that if A is any strictly upper triangular matrix in $M_n(S)$ then $A^n = 0$ (a strictly upper triangular matrix is one whose entries on and below the main diagonal are all zero).
9. Let $\alpha = r + r^2 - 2s$ and $\beta = -3r^2 + rs$ be the two elements of the integral group ring $\mathbb{Z}D_8$ described in this section. Compute the following elements of $\mathbb{Z}D_8$:
- (a) $\beta\alpha$, (b) α^2 , (c) $\alpha\beta - \beta\alpha$, (d) $\beta\alpha\beta$.
10. Consider the following elements of the integral group ring $\mathbb{Z}S_3$:

$$\alpha = 3(1\ 2) - 5(2\ 3) + 14(1\ 2\ 3) \quad \text{and} \quad \beta = 6(1) + 2(2\ 3) - 7(1\ 3\ 2)$$

- (where (1) is the identity of S_3). Compute the following elements:
- (a) $\alpha + \beta$, (b) $2\alpha - 3\beta$, (c) $\alpha\beta$, (d) $\beta\alpha$, (e) α^2 .
11. Repeat the preceding exercise under the assumption that the coefficients of α and β are in $\mathbb{Z}/3\mathbb{Z}$ (i.e., $\alpha, \beta \in \mathbb{Z}/3\mathbb{Z}S_3$).
12. Let $G = \{g_1, \dots, g_n\}$ be a finite group. Prove that the element $N = g_1 + g_2 + \dots + g_n$ is in the center of the group ring RG (cf. Exercise 7, Section 1).
13. Let $\mathcal{K} = \{k_1, \dots, k_m\}$ be a conjugacy class in the finite group G .
- (a) Prove that the element $K = k_1 + \dots + k_m$ is in the center of the group ring RG (cf. Exercise 7, Section 1). [Check that $g^{-1}Kg = K$ for all $g \in G$.]
- (b) Let $\mathcal{K}_1, \dots, \mathcal{K}_r$ be the conjugacy classes of G and for each $\mathcal{K}_i$ let K_i be the element of RG that is the sum of the members of $\mathcal{K}_i$. Prove that an element $\alpha \in RG$ is in the center of RG if and only if $\alpha = a_1K_1 + a_2K_2 + \dots + a_rK_r$ for some $a_1, a_2, \dots, a_r \in R$.

7.3 RING HOMOMORPHISMS AND QUOTIENT RINGS

A ring homomorphism is a map from one ring to another that respects the additive and multiplicative structures:

Definition. Let R and S be rings.

- (1) A *ring homomorphism* is a map $\varphi : R \rightarrow S$ satisfying
- (i) $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$ (so φ is a group homomorphism on the additive groups) and
- (ii) $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$.
- (2) The *kernel* of the ring homomorphism φ , denoted $\ker \varphi$, is the set of elements of R that map to 0 in S (i.e., the kernel of φ viewed as a homomorphism of additive groups).
- (3) A bijective ring homomorphism is called an *isomorphism*.

If the context is clear we shall simply use the term “homomorphism” instead of “ring homomorphism.” Similarly, if A and B are rings, $A \cong B$ will always mean an isomorphism of rings unless otherwise stated.

Examples

- (1) The map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ defined by sending an even integer to 0 and an odd integer to 1 is a ring homomorphism. The map is additive since the sum of two even or odd integers is even and the sum of an even integer and an odd integer is odd. The map is multiplicative since the product of two odd integers is odd and the product of an even integer with any integer is even. The kernel of φ (the fiber of φ above $0 \in \mathbb{Z}/2\mathbb{Z}$) is the set of even integers. The fiber of φ above $1 \in \mathbb{Z}/2\mathbb{Z}$ is the set of odd integers.
- (2) For $n \in \mathbb{Z}$ the maps $\varphi_n : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $\varphi_n(x) = nx$ are *not* in general ring homomorphisms because $\varphi_n(xy) = nxy$ whereas $\varphi_n(x)\varphi_n(y) = nxny = n^2xy$. Hence φ_n is a ring homomorphism only when $n^2 = n$, i.e., $n = 0, 1$. Note however that φ_n is always a *group homomorphism* on the additive groups. Thus care should be exercised when dealing with rings to be sure to check that *both* ring operations are preserved. Note that φ_0 is the zero homomorphism and φ_1 is the identity homomorphism.
- (3) Let $\varphi : \mathbb{Q}[x] \rightarrow \mathbb{Q}$ be the map from the ring of polynomials in x with rational coefficients to the rationals defined by $\varphi(p(x)) = p(0)$ (i.e., mapping the polynomial to its constant term). Then φ is a ring homomorphism since the constant term of the sum of two polynomials is the sum of their constant terms and the constant term of the product of two polynomials is the product of their constant terms. The fiber above $a \in \mathbb{Q}$ consists of the set of polynomials with a as constant term. In particular, the kernel of φ consists of the polynomials with constant term 0.

Proposition 5. Let R and S be rings and let $\varphi : R \rightarrow S$ be a homomorphism.

- (1) The image of φ is a subring of S .
- (2) The kernel of φ is a subring of R . Furthermore, if $\alpha \in \ker \varphi$ then $r\alpha$ and $\alpha r \in \ker \varphi$ for every $r \in R$, i.e., $\ker \varphi$ is closed under multiplication by elements from R .

Proof: (1) If $s_1, s_2 \in \text{im } \varphi$ then $s_1 = \varphi(r_1)$ and $s_2 = \varphi(r_2)$ for some $r_1, r_2 \in R$. Then $\varphi(r_1 - r_2) = s_1 - s_2$ and $\varphi(r_1 r_2) = s_1 s_2$. This shows $s_1 - s_2, s_1 s_2 \in \text{im } \varphi$, so the image of φ is closed under subtraction and under multiplication, hence is a subring of S .

(2) If $\alpha, \beta \in \ker \varphi$ then $\varphi(\alpha) = \varphi(\beta) = 0$. Hence $\varphi(\alpha - \beta) = 0$ and $\varphi(\alpha\beta) = 0$, so $\ker \varphi$ is closed under subtraction and under multiplication, so is a subring of R . Similarly, for any $r \in R$ we have $\varphi(r\alpha) = \varphi(r)\varphi(\alpha) = \varphi(r) 0 = 0$, and also $\varphi(\alpha r) = \varphi(\alpha)\varphi(r) = 0\varphi(r) = 0$, so $r\alpha, \alpha r \in \ker \varphi$.

In the case of a homomorphism φ of groups we saw that the fibers of the homomorphism have the structure of a group naturally isomorphic to the image of φ , which led to the notion of a quotient group by a normal subgroup. An analogous result is true for a homomorphism of rings.

Let $\varphi : R \rightarrow S$ be a ring homomorphism with kernel I . Since R and S are in particular additive abelian groups, φ is in particular a homomorphism of abelian groups

and the fibers of φ are the additive cosets $r + I$ of the kernel I (more precisely, if r is any element of R mapping to $a \in S$, $\varphi(r) = a$, then the fiber of φ over a is the coset $r + I$ of the kernel I). These fibers have the structure of a ring naturally isomorphic to the image of φ : if X is the fiber over $a \in S$ and Y is the fiber over $b \in S$, then $X + Y$ is the fiber over $a + b$ and XY is the fiber over ab . In terms of cosets of the kernel I this addition and multiplication is

$$(r + I) + (s + I) = (r + s) + I \quad (7.1)$$

$$(r + I) \times (s + I) = (rs) + I. \quad (7.2)$$

As in the case for groups, the verification that these operations define a ring structure on the collection of cosets of the kernel I ultimately rests on the corresponding ring properties of S . This ring of cosets is called the *quotient ring* of R by $I = \ker \varphi$ and is denoted R/I . Note that the additive structure of the ring R/I is just the additive quotient group of the additive abelian group R by the (necessarily normal) subgroup I . When I is the kernel of some homomorphism φ this additive abelian quotient group also has a multiplicative structure, defined by (7.2), which makes R/I into a ring.

As in the case for groups, we can also consider whether (1) and (2) can be used to define a ring structure on the collection of cosets of an *arbitrary* subgroup I of R . Note that since R is an abelian additive group, the subgroup I is necessarily normal so that the quotient R/I of cosets of I is automatically an additive abelian group. The question then is whether this quotient group also has a *multiplicative* structure induced from the multiplication in R , defined by (2). The answer is no in general (just as the answer is no in trying to form the quotient by an arbitrary subgroup of a group), which leads to the notion of an *ideal* in R (the analogue for rings of a normal subgroup of a group). We shall then see that the ideals of R are exactly the kernels of the ring homomorphisms of R (the analogue for rings of the characterization of normal subgroups as the kernels of group homomorphisms).

Let I be an arbitrary subgroup of the additive group R . We consider when the multiplication of cosets in (2) is well defined and makes the additive abelian group R/I into a ring. The statement that the multiplication in (2) is well defined is the statement that the multiplication is independent of the particular representatives r and s chosen, i.e., that we obtain the same coset on the right if instead we use the representatives $r + \alpha$ and $s + \beta$ for any $\alpha, \beta \in I$. In other words, we must have

$$(r + \alpha)(s + \beta) + I = rs + I \quad (*)$$

for all $r, s \in R$ and all $\alpha, \beta \in I$.

Letting $r = s = 0$, we see that I must be closed under multiplication, i.e., I must be a *subring* of R .

Next, by letting $s = 0$ and letting r be arbitrary, we see that we must have $r\beta \in I$ for every $r \in R$ and every $\beta \in I$, i.e., that I must be closed under multiplication on the left by elements from R . Letting $r = 0$ and letting s be arbitrary, we see similarly that I must be closed under multiplication on the right by elements from R .

Conversely, if I is closed under multiplication on the left and on the right by elements from R then the relation $(*)$ is satisfied for all $\alpha, \beta \in I$. Hence this is a necessary and sufficient condition for the multiplication in (2) to be well defined.

Finally, if the multiplication of cosets defined by (2) is well defined, then this multiplication makes the additive quotient group R/I into a ring. Each ring axiom in the quotient follows directly from the corresponding axiom in R . For example, one of the distributive laws is verified as follows:

$$\begin{aligned}(r + I)[(s + I) + (t + I)] &= (r + I)[(s + t) + I] \\ &= r(s + t) + I = (rs + rt) + I \\ &= (rs + I) + (rt + I) \\ &= [(r + I)(s + I)] + [(r + I)(t + I)].\end{aligned}$$

This shows that the quotient R/I of the ring R by a subgroup I has a natural ring structure if and only if I is also closed under multiplication on the left and on the right by elements from R (so in particular must be a subring of R since it is closed under multiplication). As mentioned, such subrings I are called the *ideals* of R :

Definition. Let R be a ring, let I be a subset of R and let $r \in R$.

- (1) $rI = \{ra \mid a \in I\}$ and $Ir = \{ar \mid a \in I\}$.
- (2) A subset I of R is a *left ideal* of R if
 - (i) I is a subring of R , and
 - (ii) I is closed under left multiplication by elements from R , i.e., $rI \subseteq I$ for all $r \in R$.
 Similarly I is a *right ideal* if (i) holds and in place of (ii) one has
 - (ii)' I is closed under right multiplication by elements from R , i.e., $Ir \subseteq I$ for all $r \in R$.
- (3) A subset I that is both a left ideal and a right ideal is called an *ideal* (or, for added emphasis, a *two-sided ideal*) of R .

For commutative rings the notions of left, right and two-sided ideal coincide. We emphasize that to prove a subset I of a ring R is an ideal it is necessary to prove that I is nonempty, closed under subtraction and closed under multiplication by all the elements of R (and not just by elements of I). If R has a 1 then $(-1)a = -a$ so in this case I is an ideal if it is nonempty, closed under addition and closed under multiplication by all the elements of R .

Note also that the last part of Proposition 5 proves that the kernel of any ring homomorphism is an ideal.

We summarize the preceding discussion in the following proposition.

Proposition 6. Let R be a ring and let I be an ideal of R . Then the (additive) quotient group R/I is a ring under the binary operations:

$$(r + I) + (s + I) = (r + s) + I \quad \text{and} \quad (r + I) \times (s + I) = (rs) + I$$

for all $r, s \in R$. Conversely, if I is any subgroup such that the above operations are well defined, then I is an ideal of R .

Definition. When I is an ideal of R the ring R/I with the operations in the previous proposition is called the *quotient ring* of R by I .

Theorem 7.

- (1) (*The First Isomorphism Theorem for Rings*) If $\varphi : R \rightarrow S$ is a homomorphism of rings, then the kernel of φ is an ideal of R , the image of φ is a subring of S and $R/\ker \varphi$ is isomorphic as a ring to $\varphi(R)$.
- (2) If I is any ideal of R , then the map

$$R \rightarrow R/I \quad \text{defined by} \quad r \mapsto r + I$$

is a surjective ring homomorphism with kernel I (this homomorphism is called the *natural projection* of R onto R/I). Thus every ideal is the kernel of a ring homomorphism and vice versa.

Proof: This is just a matter of collecting previous calculations. If I is the kernel of φ , then the cosets (under addition) of I are precisely the fibers of φ . In particular, the cosets $r + I$, $s + I$ and $rs + I$ are the fibers of φ over $\varphi(r)$, $\varphi(s)$ and $\varphi(rs)$, respectively. Since φ is a ring homomorphism $\varphi(r)\varphi(s) = \varphi(rs)$, hence $(r + I)(s + I) = rs + I$. Multiplication of cosets is well defined and so I is an ideal and R/I is a ring. The correspondence $r + I \mapsto \varphi(r)$ is a bijection between the rings R/I and $\varphi(R)$ which respects addition and multiplication, hence is a ring isomorphism.

If I is any ideal, then R/I is a ring (in particular is an abelian group) and the map $\pi : r \mapsto r + I$ is a group homomorphism with kernel I . It remains to check that π is a ring homomorphism. This is immediate from the definition of multiplication in R/I :

$$\pi : rs \mapsto rs + I = (r + I)(s + I) = \pi(r)\pi(s).$$

As with groups we shall often use the bar notation for reduction mod I : $\bar{r} = r + I$. With this notation the addition and multiplication in the quotient ring R/I become simply $\bar{r} + \bar{s} = \overline{r + s}$ and $\bar{r}\bar{s} = \overline{rs}$.

Examples

Let R be a ring.

- (1) The subrings R and $\{0\}$ are ideals. An ideal I is *proper* if $I \neq R$. The ideal $\{0\}$ is called the *trivial ideal* and is denoted by 0 .
- (2) It is immediate that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ for any $n \in \mathbb{Z}$ and these are the only ideals of $\mathbb{Z}$ since in particular these are the only subgroups of $\mathbb{Z}$. The associated quotient ring is $\mathbb{Z}/n\mathbb{Z}$ (which explains the choice of notation and which we have now proved is a ring), introduced in Chapter 0. For example, if $n = 15$ then the elements of $\mathbb{Z}/15\mathbb{Z}$ are the cosets $\bar{0}, \bar{1}, \dots, \bar{13}, \bar{14}$. To add (or multiply) in the quotient, simply choose any representatives for the two cosets, add (multiply, respectively) these representatives in the integers $\mathbb{Z}$, and take the corresponding coset containing this sum (product, respectively). For example, $\bar{7} + \bar{11} = \bar{18}$ and $\bar{18} = \bar{3}$, so $\bar{7} + \bar{11} = \bar{3}$ in $\mathbb{Z}/15\mathbb{Z}$. Similarly, $\bar{7}\bar{11} = \bar{77} = \bar{2}$ in $\mathbb{Z}/15\mathbb{Z}$. We could also express this by writing $7 + 11 \equiv 3 \pmod{15}$, $7(11) \equiv 2 \pmod{15}$.

The natural projection $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is called *reduction mod n* and will be discussed further at the end of these examples.

- (3) Let $R = \mathbb{Z}[x]$ be the ring of polynomials in x with integer coefficients. Let I be the collection of polynomials whose terms are of degree at least 2 (i.e., having no terms of degree 0 or degree 1) together with the zero polynomial. Then I is an ideal: the sum of two such polynomials again has terms of degree at least 2 and the product of a polynomial whose terms are of degree at least 2 with *any* polynomial again only has terms of degree at least 2. Two polynomials $p(x), q(x)$ are in the same coset of I if and only if they differ by a polynomial whose terms are of degree at least 2, i.e., if and only if $p(x)$ and $q(x)$ have the same constant and first degree terms. For example, the polynomials $3 + 5x + x^3 + x^5$ and $3 + 5x - x^4$ are in the same coset of I . It follows easily that a complete set of representatives for the quotient R/I is given by the polynomials $a + bx$ of degree at most 1.

Addition and multiplication in the quotient are again performed by representatives. For example,

$$(\overline{1 + 3x}) + (\overline{-4 + 5x}) = \overline{-3 + 8x}$$

and

$$(\overline{1 + 3x})(\overline{-4 + 5x}) = \overline{(-4 - 7x + 15x^2)} = \overline{-4 - 7x}.$$

Note that in this quotient ring R/I we have $\bar{x} \bar{x} = \overline{x^2} = \bar{0}$, for example, so that R/I has zero divisors, even though $R = \mathbb{Z}[x]$ does not.

- (4) Let A be a ring, let X be any nonempty set and let R be the ring of all functions from X to A . For each fixed $c \in X$ the map

$$E_c : R \rightarrow A \quad \text{defined by} \quad E_c(f) = f(c)$$

(called *evaluation at c*) is a ring homomorphism because the operations in R are pointwise addition and multiplication of functions. The kernel of E_c is given by $\{f \in R \mid f(c) = 0\}$ (the set of functions from X to A that vanish at c). Also, E_c is surjective: given any $a \in A$ the constant function $f(x) = a$ maps to a under evaluation at c . Thus $R/\ker E_c \cong A$.

Similarly, let X be the closed interval $[0,1]$ in $\mathbb{R}$ and let R be the ring of all continuous real valued functions on $[0,1]$. For each $c \in [0,1]$, evaluation at c is a surjective ring homomorphism (since R contains the constant functions) and so $R/\ker E_c \cong \mathbb{R}$. The kernel of E_c is the ideal of all continuous functions whose graph crosses the x -axis at c . More generally, the fiber of E_c above the real number y_0 is the set of all continuous functions that pass through the point (c, y_0) .

- (5) The map from the polynomial ring $R[x]$ to R defined by $p(x) \mapsto p(0)$ (evaluation at 0) is a ring homomorphism whose kernel is the set of all polynomials whose constant term is zero, i.e., $p(0) = 0$. We can compose this homomorphism with any homomorphism from R to another ring S to obtain a ring homomorphism from $R[x]$ to S . For example, let $R = \mathbb{Z}$ and consider the homomorphism $\mathbb{Z}[x] \rightarrow \mathbb{Z}/2\mathbb{Z}$ defined by the composition $p(x) \mapsto p(0) \mapsto p(0) \bmod 2 \in \mathbb{Z}/2\mathbb{Z}$. The kernel of this composite map is given by $\{p(x) \in \mathbb{Z}[x] \mid p(0) \in 2\mathbb{Z}\}$, i.e., the set of all polynomials with integer coefficients whose constant term is even. The other fiber of this homomorphism is the coset of polynomials whose constant term is odd, as we determined earlier. Since the homomorphism is clearly surjective, the quotient ring is $\mathbb{Z}/2\mathbb{Z}$.
- (6) Fix some $n \in \mathbb{Z}$ with $n \geq 2$ and consider the noncommutative ring $M_n(R)$. If J is any ideal of R then $M_n(J)$, the $n \times n$ matrices whose entries come from J , is a two-sided ideal of $M_n(R)$. This ideal is the kernel of the surjective homomorphism $M_n(R) \rightarrow M_n(R/J)$ which reduces each entry of a matrix mod J , i.e., which maps each entry a_{ij} to $\bar{a}_{ij}$ (here bar denotes passage to R/J). For instance, when $n = 3$ and $R = \mathbb{Z}$, the 3×3 matrices whose entries are all even is the two-sided ideal $M_3(2\mathbb{Z})$.

of $M_3(\mathbb{Z})$ and the quotient $M_3(\mathbb{Z})/M_3(2\mathbb{Z})$ is isomorphic to $M_3(\mathbb{Z}/2\mathbb{Z})$. If the ring R has an identity then the exercises below show that every two-sided ideal of $M_n(R)$ is of the form $M_n(J)$ for some two-sided ideal J of R .

- (7) Let R be a commutative ring with 1 and let $G = \{g_1, \dots, g_n\}$ be a finite group. The map from the group ring RG to R defined by $\sum_{i=1}^n a_i g_i \mapsto \sum_{i=1}^n a_i$ is easily seen to be a homomorphism, called the *augmentation map*. The kernel of the augmentation map, the *augmentation ideal*, is the set of elements of RG whose coefficients sum to 0. For example, $g_i - g_j$ is an element of the augmentation ideal for all i, j . Since the augmentation map is surjective, the quotient ring is isomorphic to R .

Another ideal in RG is $\{\sum_{i=1}^n a g_i \mid a \in R\}$, i.e., the formal sums whose coefficients are all equal (equivalently, all R -multiples of the element $g_1 + \dots + g_n$).

- (8) Let R be a commutative ring with identity $1 \neq 0$ and let $n \in \mathbb{Z}$ with $n \geq 2$. We exhibit some one-sided ideals in the ring $M_n(R)$. For each $j \in \{1, 2, \dots, n\}$ let L_j be the set of all $n \times n$ matrices in $M_n(R)$ with arbitrary entries in the j^{th} column and zeros in all other columns. It is clear that L_j is closed under subtraction. It follows directly from the definition of matrix multiplication that for any matrix $T \in M_n(R)$ and any $A \in L_j$ the product TA has zero entries in the i^{th} column for all $i \neq j$. This shows L_j is a *left ideal* of $M_n(R)$. Moreover, L_j is *not* a *right ideal* (hence is not a two-sided ideal). To see this, let E_{pq} be the matrix with 1 in the p^{th} row and q^{th} column and zeros elsewhere ($p, q \in \{1, \dots, n\}$). Then $E_{1j} \in L_j$ but $E_{1j}E_{ji} = E_{1i} \notin L_j$ if $i \neq j$, so L_j is not closed under right multiplication by arbitrary ring elements. An analogous argument shows that if R_j is the set of all $n \times n$ matrices in $M_n(R)$ with arbitrary entries in the j^{th} row and zeros in all other rows, then R_j is a *right ideal* which is not a *left ideal*. These one-sided ideals will play an important role in Part VI.

Example: (The Reduction Homomorphism)

The canonical projection map from $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$ obtained by factoring out by the ideal $n\mathbb{Z}$ of $\mathbb{Z}$ is usually referred to as “reducing modulo n .” The fact that this is a *ring homomorphism* has important consequences for elementary number theory. For example, suppose we are trying to solve the equation

$$x^2 + y^2 = 3z^2$$

in *integers* x, y and z (such problems are frequently referred to as *Diophantine equations* after Diophantus, who was one of the first to systematically examine the existence of *integer* solutions of equations). Suppose such integers exist. Observe first that we may assume x, y and z have no factors in common, since otherwise we could divide through this equation by the square of this common factor and obtain another set of integer solutions smaller than the initial ones. This equation simply states a relation between these elements in the *ring* $\mathbb{Z}$. As such, the same relation must also hold in any *quotient* ring as well. In particular, this relation must hold in $\mathbb{Z}/n\mathbb{Z}$ for any integer n . The choice $n = 4$ is particularly efficacious, for the following reason: the squares mod 4 are just $0^2, 1^2, 2^2, 3^2$, i.e., $0, 1 \pmod{4}$. Reading the above equation mod 4 (that is, considering this equation in the quotient ring $\mathbb{Z}/4\mathbb{Z}$), we must have

$$\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \equiv 3 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \equiv \begin{Bmatrix} 0 \\ 3 \end{Bmatrix} \pmod{4}$$

where the $\begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$, for example, indicates that either a 0 or a 1 may be taken. Checking the few possibilities shows that we must take the 0 each time. This means that each

of x , y and z must be even integers (squares of the odd integers gave us $1 \bmod 4$). But this contradicts the assumption of no common factors for these integers, and shows that this equation has *no solutions in nonzero integers*.

Note that even had solutions existed, this technique gives information about the possible residues of the solutions $\bmod n$ (since we could just as well have examined the possibilities $\bmod n$ as $\bmod 4$) and note that for each choice of n we have only a *finite* problem to solve because there are only finitely many residue classes $\bmod n$. Together with the Chinese Remainder Theorem (described in Section 6), we can then determine the possible solutions modulo very large integers, which greatly assists in finding them numerically (when they exist). We also observe that this technique has a number of limitations — for example, there are equations which have solutions modulo every integer, but which do not have integer solutions. An easy example (but extremely hard to verify that it does indeed have this property) is the equation

$$3x^3 + 4y^3 + 5z^3 = 0.$$

As a final example of this technique, we mention that the map from the ring $\mathbb{Z}[x]$ of polynomials with integer coefficients to the ring $\mathbb{Z}/p\mathbb{Z}[x]$ of polynomials with coefficients in $\mathbb{Z}/p\mathbb{Z}$ for a prime p given by *reducing the coefficients modulo p* is a ring homomorphism. This example of reduction will be used in Chapter 9 in trying to determine whether polynomials can be factored.

The following theorem gives the remaining Isomorphism Theorems for rings. Each of these may be proved as follows: first use the corresponding theorem from group theory to obtain an isomorphism of *additive groups* (or correspondence of groups, in the case of the Fourth Isomorphism Theorem) and then check that this group isomorphism (or correspondence, respectively) is a multiplicative map, and so defines a *ring* isomorphism. In each case the verification is immediate from the definition of multiplication in quotient rings. For example, the map that gives the isomorphism in (2) below is defined by $\varphi : r + I \mapsto r + J$. This map is multiplicative since $(r_1 + I)(r_2 + I) = r_1r_2 + I$ by the definition of the multiplication in the quotient ring R/I , and $r_1r_2 + I \mapsto r_1r_2 + J = (r_1 + J)(r_2 + J)$ by the definition of the multiplication in the quotient ring R/J , i.e., $\varphi(r_1r_2) = \varphi(r_1)\varphi(r_2)$. The proofs for the other parts of the theorem are similar.

Theorem 8. Let R be a ring.

- (1) (*The Second Isomorphism Theorem for Rings*) Let A be a subring and let B be an ideal of R . Then $A + B = \{a + b \mid a \in A, b \in B\}$ is a subring of R , $A \cap B$ is an ideal of A and $(A + B)/B \cong A/(A \cap B)$.
- (2) (*The Third Isomorphism Theorem for Rings*) Let I and J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and $(R/I)/(J/I) \cong R/J$.
- (3) (*The Fourth or Lattice Isomorphism Theorem for Rings*) Let I be an ideal of R . The correspondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings A of R that contain I and the set of subrings of R/I . Furthermore, A (a subring containing I) is an ideal of R if and only if A/I is an ideal of R/I .

Let $R = \mathbb{Z}$ and let I be the ideal $12\mathbb{Z}$. The quotient ring $\bar{R} = R/I = \mathbb{Z}/12\mathbb{Z}$ has ideals $\bar{R}$, $2\mathbb{Z}/12\mathbb{Z}$, $3\mathbb{Z}/12\mathbb{Z}$, $4\mathbb{Z}/12\mathbb{Z}$, $6\mathbb{Z}/12\mathbb{Z}$, and $\bar{0} = 12\mathbb{Z}/12\mathbb{Z}$ corresponding to the ideals $R = \mathbb{Z}$, $2\mathbb{Z}$, $3\mathbb{Z}$, $4\mathbb{Z}$, $6\mathbb{Z}$ and $12\mathbb{Z} = I$ of R containing I , respectively.

If I and J are ideals in the ring R then the set of sums $a + b$ with $a \in I$ and $b \in J$ is not only a subring of R (as in the Second Isomorphism Theorem for Rings), but is an *ideal* in R (the set is clearly closed under sums and $r(a + b) = ra + rb \in I + J$ since $ra \in I$ and $rb \in J$). We can also define the product of two ideals:

Definition. Let I and J be ideals of R .

- (1) Define the *sum* of I and J by $I + J = \{a + b \mid a \in I, b \in J\}$.
- (2) Define the *product* of I and J , denoted by IJ , to be the set of all finite sums of elements of the form ab with $a \in I$ and $b \in J$.
- (3) For any $n \geq 1$, define the n^{th} *power* of I , denoted by I^n , to be the set consisting of all finite sums of elements of the form $a_1 a_2 \cdots a_n$ with $a_i \in I$ for all i . Equivalently, I^n is defined inductively by defining $I^1 = I$, and $I^n = I I^{n-1}$ for $n = 2, 3, \dots$.

It is easy to see that the sum $I + J$ of the ideals I and J is the smallest ideal of R containing both I and J and that the product IJ is an ideal contained in $I \cap J$ (but may be strictly smaller, cf. the exercises). Note also that the elements of the product ideal IJ are *finite sums* of products of elements ab from I and J . The set $\{ab \mid a \in I, b \in J\}$ consisting just of products of elements from I and J is in general not closed under addition, hence is not in general an ideal.

Examples

- (1) Let $I = 6\mathbb{Z}$ and $J = 10\mathbb{Z}$ in $\mathbb{Z}$. Then $I + J$ consists of all integers of the form $6x + 10y$ with $x, y \in \mathbb{Z}$. Since every such integer is divisible by 2, the ideal $I + J$ is contained in $2\mathbb{Z}$. On the other hand, $2 = 6(2) + 10(-1)$ shows that the ideal $I + J$ contains the ideal $2\mathbb{Z}$, so that $6\mathbb{Z} + 10\mathbb{Z} = 2\mathbb{Z}$. In general, $m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}$, where d is the greatest common divisor of m and n . The product IJ consists of all finite sums of elements of the form $(6x)(10y)$ with $x, y \in \mathbb{Z}$, which clearly gives the ideal $60\mathbb{Z}$.
- (2) Let I be the ideal in $\mathbb{Z}[x]$ consisting of the polynomials with integer coefficients whose constant term is even (cf. Example 5). The two polynomials 2 and x are contained in I , so both $4 = 2 \cdot 2$ and $x^2 = x \cdot x$ are elements of the product ideal $I^2 = II$, as is their sum $x^2 + 4$. It is easy to check, however, that $x^2 + 4$ cannot be written as a single product $p(x)q(x)$ of two elements of I .

EXERCISES

Let R be a ring with identity $1 \neq 0$.

1. Prove that the rings $2\mathbb{Z}$ and $3\mathbb{Z}$ are not isomorphic.
2. Prove that the rings $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ are not isomorphic.
3. Find all homomorphic images of $\mathbb{Z}$.

4. Find all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}/30\mathbb{Z}$. In each case describe the kernel and the image.
5. Describe all ring homomorphisms from the ring $\mathbb{Z} \times \mathbb{Z}$ to $\mathbb{Z}$. In each case describe the kernel and the image.
6. Decide which of the following are ring homomorphisms from $M_2(\mathbb{Z})$ to $\mathbb{Z}$:
- (a) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a$ (projection onto the 1,1 entry)
 - (b) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a + d$ (the *trace* of the matrix)
 - (c) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto ad - bc$ (the *determinant* of the matrix).
7. Let $R = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid a, b, d \in \mathbb{Z} \right\}$ be the subring of $M_2(\mathbb{Z})$ of upper triangular matrices. Prove that the map

$$\varphi : R \rightarrow \mathbb{Z} \times \mathbb{Z} \quad \text{defined by} \quad \varphi : \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto (a, d)$$

is a surjective homomorphism and describe its kernel.

8. Decide which of the following are ideals of the ring $\mathbb{Z} \times \mathbb{Z}$:
- (a) $\{(a, a) \mid a \in \mathbb{Z}\}$
 - (b) $\{(2a, 2b) \mid a, b \in \mathbb{Z}\}$
 - (c) $\{(2a, 0) \mid a \in \mathbb{Z}\}$
 - (d) $\{(a, -a) \mid a \in \mathbb{Z}\}$.
9. Decide which of the sets in Exercise 6 of Section 1 are ideals of the ring of all functions from $[0,1]$ to $\mathbb{R}$.
10. Decide which of the following are ideals of the ring $\mathbb{Z}[x]$:
- (a) the set of all polynomials whose constant term is a multiple of 3
 - (b) the set of all polynomials whose coefficient of x^2 is a multiple of 3
 - (c) the set of all polynomials whose constant term, coefficient of x and coefficient of x^2 are zero
 - (d) $\mathbb{Z}[x^2]$ (i.e., the polynomials in which only even powers of x appear)
 - (e) the set of polynomials whose coefficients sum to zero
 - (f) the set of polynomials $p(x)$ such that $p'(0) = 0$, where $p'(x)$ is the usual first derivative of $p(x)$ with respect to x .
11. Let R be the ring of all continuous real valued functions on the closed interval $[0, 1]$. Prove that the map $\varphi : R \rightarrow \mathbb{R}$ defined by $\varphi(f) = \int_0^1 f(t)dt$ is a homomorphism of additive groups but not a ring homomorphism.
12. Let D be an integer that is not a perfect square in $\mathbb{Z}$ and let $S = \left\{ \begin{pmatrix} a & b \\ Db & a \end{pmatrix} \mid a, b \in \mathbb{Z} \right\}$.
- (a) Prove that S is a subring of $M_2(\mathbb{Z})$.
 - (b) If D is not a perfect square in $\mathbb{Z}$ prove that the map $\varphi : \mathbb{Z}[\sqrt{D}] \rightarrow S$ defined by $\varphi(a + b\sqrt{D}) = \begin{pmatrix} a & b \\ Db & a \end{pmatrix}$ is a ring isomorphism.
 - (c) If $D \equiv 1 \pmod{4}$ is squarefree, prove that the set $\left\{ \begin{pmatrix} a & b \\ (D-1)b/4 & a+b \end{pmatrix} \mid a, b \in \mathbb{Z} \right\}$

is a subring of $M_2(\mathbb{Z})$ and is isomorphic to the quadratic integer ring $\mathcal{O}$.

13. Prove that the ring $M_2(\mathbb{R})$ contains a subring that is isomorphic to $\mathbb{C}$.
14. Prove that the ring $M_4(\mathbb{R})$ contains a subring that is isomorphic to the real Hamilton Quaternions, $\mathbb{H}$.
15. Let X be a nonempty set and let $\mathcal{P}(X)$ be the Boolean ring of all subsets of X defined in Exercise 21 of Section 1. Let R be the ring of all functions from X into $\mathbb{Z}/2\mathbb{Z}$. For each $A \in \mathcal{P}(X)$ define the function

$$\chi_A : X \rightarrow \mathbb{Z}/2\mathbb{Z} \quad \text{by} \quad \chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

- (χ_A is called the *characteristic function* of A with values in $\mathbb{Z}/2\mathbb{Z}$). Prove that the map $\mathcal{P}(X) \rightarrow R$ defined by $A \mapsto \chi_A$ is a ring isomorphism.
16. Let $\varphi : R \rightarrow S$ be a surjective homomorphism of rings. Prove that the image of the center of R is contained in the center of S (cf. Exercise 7 of Section 1).
17. Let R and S be nonzero rings with identity and denote their respective identities by 1_R and 1_S . Let $\varphi : R \rightarrow S$ be a nonzero homomorphism of rings.
- (a) Prove that if $\varphi(1_R) \neq 1_S$ then $\varphi(1_R)$ is a zero divisor in S . Deduce that if S is an integral domain then every ring homomorphism from R to S sends the identity of R to the identity of S .
- (b) Prove that if $\varphi(1_R) = 1_S$ then $\varphi(u)$ is a unit in S and that $\varphi(u^{-1}) = \varphi(u)^{-1}$ for each unit u of R .
18. (a) If I and J are ideals of R prove that their intersection $I \cap J$ is also an ideal of R .
 (b) Prove that the intersection of an arbitrary nonempty collection of ideals is again an ideal (do not assume the collection is countable).
19. Prove that if $I_1 \subseteq I_2 \subseteq \cdots$ are ideals of R then $\bigcup_{n=1}^{\infty} I_n$ is an ideal of R .
20. Let I be an ideal of R and let S be a subring of R . Prove that $I \cap S$ is an ideal of S . Show by example that not every ideal of a subring S of a ring R need be of the form $I \cap S$ for some ideal I of R .
21. Prove that every (two-sided) ideal of $M_n(R)$ is equal to $M_n(J)$ for some (two-sided) ideal J of R . [Use Exercise 6(c) of Section 2 to show first that the set of entries of matrices in an ideal of $M_n(R)$ form an ideal in R .]
22. Let a be an element of the ring R .
- (a) Prove that $\{x \in R \mid ax = 0\}$ is a right ideal and $\{y \in R \mid ya = 0\}$ is a left ideal (called respectively the right and left *annihilators* of a in R).
- (b) Prove that if L is a left ideal of R then $\{x \in R \mid xa = 0 \text{ for all } a \in L\}$ is a two-sided ideal (called the left *annihilator* of L in R).
23. Let S be a subring of R and let I be an ideal of R . Prove that if $S \cap I = 0$ then $\bar{S} \cong S$, where the bar denotes passage to R/I .
24. Let $\varphi : R \rightarrow S$ be a ring homomorphism.
- (a) Prove that if J is an ideal of S then $\varphi^{-1}(J)$ is an ideal of R . Apply this to the special case when R is a subring of S and φ is the inclusion homomorphism to deduce that if J is an ideal of S then $J \cap R$ is an ideal of R .
- (b) Prove that if φ is surjective and I is an ideal of R then $\varphi(I)$ is an ideal of S . Give an example where this fails if φ is not surjective.
25. Assume R is a commutative ring with 1. Prove that the Binomial Theorem

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

holds in R , where the binomial coefficient $\binom{n}{k}$ is interpreted in R as the sum $1 + 1 + \cdots + 1$ of the identity 1 in R taken $\binom{n}{k}$ times.

26. The *characteristic* of a ring R is the smallest positive integer n such that $1 + 1 + \cdots + 1 = 0$ (n times) in R ; if no such integer exists the characteristic of R is said to be 0. For example, $\mathbb{Z}/n\mathbb{Z}$ is a ring of characteristic n for each positive integer n and $\mathbb{Z}$ is a ring of characteristic 0.

(a) Prove that the map $\mathbb{Z} \rightarrow R$ defined by

$$k \mapsto \begin{cases} 1 + 1 + \cdots + 1 \text{ (} k \text{ times)} & \text{if } k > 0 \\ 0 & \text{if } k = 0 \\ -1 - 1 - \cdots - 1 \text{ (} -k \text{ times)} & \text{if } k < 0 \end{cases}$$

is a ring homomorphism whose kernel is $n\mathbb{Z}$, where n is the characteristic of R (this explains the use of the terminology “characteristic 0” instead of the archaic phrase “characteristic ∞ ” for rings in which no sum of 1’s is zero).

(b) Determine the characteristics of the rings $\mathbb{Q}$, $\mathbb{Z}[x]$, $\mathbb{Z}/n\mathbb{Z}[x]$.

(c) Prove that if p is a prime and if R is a commutative ring of characteristic p , then $(a + b)^p = a^p + b^p$ for all $a, b \in R$.

27. Prove that a nonzero Boolean ring has characteristic 2 (cf. Exercise 15, Section 1).
28. Prove that an integral domain has characteristic p , where p is either a prime or 0 (cf. Exercise 26).
29. Let R be a commutative ring. Recall (cf. Exercise 13, Section 1) that an element $x \in R$ is nilpotent if $x^n = 0$ for some $n \in \mathbb{Z}^+$. Prove that the set of nilpotent elements form an ideal — called the *nilradical* of R and denoted by $\mathfrak{N}(R)$. [Use the Binomial Theorem to show $\mathfrak{N}(R)$ is closed under addition.]
30. Prove that if R is a commutative ring and $\mathfrak{N}(R)$ is its nilradical (cf. the preceding exercise) then zero is the only nilpotent element of $R/\mathfrak{N}(R)$ i.e., prove that $\mathfrak{N}(R/\mathfrak{N}(R)) = 0$.
31. Prove that the elements $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ are nilpotent elements of $M_2(\mathbb{Z})$ whose sum is not nilpotent (note that these two matrices do not commute). Deduce that the set of nilpotent elements in the noncommutative ring $M_2(\mathbb{Z})$ is not an ideal.
32. Let $\varphi : R \rightarrow S$ be a homomorphism of rings. Prove that if x is a nilpotent element of R then $\varphi(x)$ is nilpotent in S .
33. Assume R is commutative. Let $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$.
- (a) Prove that $p(x)$ is a unit in $R[x]$ if and only if a_0 is a unit and $a_1, a_2, \dots, a_n$ are nilpotent in R . [See Exercise 14 of Section 1.]
- (b) Prove that $p(x)$ is nilpotent in $R[x]$ if and only if $a_0, a_1, \dots, a_n$ are nilpotent elements of R .
34. Let I and J be ideals of R .
- (a) Prove that $I + J$ is the smallest ideal of R containing both I and J .
- (b) Prove that IJ is an ideal contained in $I \cap J$.
- (c) Give an example where $IJ \neq I \cap J$.
- (d) Prove that if R is commutative and if $I + J = R$ then $IJ = I \cap J$.
35. Let I, J, K be ideals of R .
- (a) Prove that $I(J + K) = IJ + IK$ and $(I + J)K = IK + JK$.
- (b) Prove that if $J \subseteq I$ then $I \cap (J + K) = J + (I \cap K)$.

36. Show that if I is the ideal of all polynomials in $\mathbb{Z}[x]$ with zero constant term then $I^n = \{a_n x^n + a_{n+1} x^{n+1} + \cdots + a_{n+m} x^{n+m} \mid a_i \in \mathbb{Z}, m \geq 0\}$ is the set of polynomials whose first nonzero term has degree at least n .
37. An ideal N is called *nilpotent* if N^n is the zero ideal for some $n \geq 1$. Prove that the ideal $p\mathbb{Z}/p^m\mathbb{Z}$ is a nilpotent ideal in the ring $\mathbb{Z}/p^m\mathbb{Z}$.

7.4 PROPERTIES OF IDEALS

Throughout this section R is a ring with identity $1 \neq 0$.

Definition. Let A be any subset of the ring R .

- (1) Let (A) denote the smallest ideal of R containing A , called *the ideal generated by A* .
- (2) Let RA denote the set of all finite sums of elements of the form ra with $r \in R$ and $a \in A$ i.e., $RA = \{r_1 a_1 + r_2 a_2 + \cdots + r_n a_n \mid r_i \in R, a_i \in A, n \in \mathbb{Z}^+\}$ (where the convention is $RA = 0$ if $A = \emptyset$).
Similarly, $AR = \{a_1 r_1 + a_2 r_2 + \cdots + a_n r_n \mid r_i \in R, a_i \in A, n \in \mathbb{Z}^+\}$ and $RAR = \{r_1 a_1 r'_1 + r_2 a_2 r'_2 + \cdots + r_n a_n r'_n \mid r_i, r'_i \in R, a_i \in A, n \in \mathbb{Z}^+\}$.
- (3) An ideal generated by a single element is called a *principal ideal*.
- (4) An ideal generated by a finite set is called a *finitely generated ideal*.

When $A = \{a\}$ or $\{a_1, a_2, \dots\}$, etc., we shall drop the set brackets and simply write (a) , $(a_1, a_2, \dots)$ for (A) , respectively.

The notion of ideals generated by subsets of a ring is analogous to that of subgroups generated by subsets of a group (Section 2.4). Since the intersection of any nonempty collection of ideals of R is also an ideal (cf. Exercise 18, Section 3) and A is always contained in at least one ideal (namely R), we have

$$(A) = \bigcap_{\substack{I \text{ an ideal} \\ A \subseteq I}} I,$$

i.e., (A) is the intersection of all ideals of R that contain the set A .

The *left ideal generated by A* is the intersection of all left ideals of R that contain A . This left ideal is obtained from A by closing A under all the operations that define a left ideal. It is immediate from the definition that RA is closed under addition and under left multiplication by any ring element. Since R has an identity, RA contains A . Thus RA is a left ideal of R which contains A . Conversely, any left ideal which contains A must contain all finite sums of elements of the form ra , $r \in R$ and $a \in A$ and so must contain RA . Thus RA is *precisely the left ideal generated by A* . Similarly, AR is *the right ideal generated by A* and RAR is *the (two-sided) ideal generated by A* . In particular,

if R is commutative then $RA = AR = RAR = (A)$.

When R is a commutative ring and $a \in R$, the principal ideal (a) generated by a is just the set of all R -multiples of a . If R is not commutative, however, the set

$\{ras \mid r, s \in R\}$ is not necessarily the two-sided ideal generated by a since it need not be closed under addition (in this case the ideal generated by a is the ideal RaR , which consists of all *finite sums* of elements of the form ras , $r, s \in R$).

The formation of principal ideals in a commutative ring is a particularly simple way of creating ideals, similar to generating cyclic subgroups of a group. Notice that the element $b \in R$ belongs to the ideal (a) if and only if $b = ra$ for some $r \in R$, i.e., if and only if b is a *multiple of a* or, put another way, a *divides b in R* . Also, $b \in (a)$ if and only if $(b) \subseteq (a)$. Thus containment relations between ideals, in particular between principal ideals, is seen to capture some of the arithmetic of general commutative rings. Commutative rings in which all ideals are principal are among the easiest to study and these will play an important role in Chapters 8 and 9.

Examples

- (1) The trivial ideal 0 and the ideal R are both principal: $0 = (0)$ and $R = (1)$.
- (2) In $\mathbb{Z}$ we have $n\mathbb{Z} = \mathbb{Z}n = (n) = (-n)$ for all integers n . Thus our notation for aR is consistent with the definition of $n\mathbb{Z}$ we have been using. As noted in the preceding section, these are all the ideals of $\mathbb{Z}$ so *every* ideal of $\mathbb{Z}$ is principal. For positive integers n and m , $n\mathbb{Z} \subseteq m\mathbb{Z}$ if and only if m divides n in $\mathbb{Z}$, so the lattice of ideals containing $n\mathbb{Z}$ is the same as the lattice of divisors of n . Furthermore, the ideal generated by two nonzero integers n and m is the principal ideal generated by their greatest common divisor, d : $(n, m) = (d)$. The notation for (n, m) as the greatest common divisor of n and m is thus consistent with the same notation for the ideal generated by n and m (although a principal generator for the ideal generated by n and m is determined only up to a $\pm$ sign — we could make it unique by choosing a nonnegative generator). In particular, n and m are relatively prime if and only if $(n, m) = (1)$.
- (3) We show that the ideal $(2, x)$ generated by 2 and x in $\mathbb{Z}[x]$ is *not* a principal ideal. Observe that $(2, x) = \{2p(x) + xq(x) \mid p(x), q(x) \in \mathbb{Z}[x]\}$ and so this ideal consists precisely of the polynomials with integer coefficients whose constant term is even (as discussed in Example 5 in the preceding section) — in particular, this is a proper ideal. Assume by way of contradiction that $(2, x) = (a(x))$ for some $a(x) \in \mathbb{Z}[x]$. Since $2 \in (a(x))$ there must be some $p(x)$ such that $2 = p(x)a(x)$. The degree of $p(x)a(x)$ equals $\deg p(x) + \deg a(x)$, hence both $p(x)$ and $a(x)$ must be constant polynomials, i.e., integers. Since 2 is a prime number, $a(x), p(x) \in \{\pm 1, \pm 2\}$. If $a(x)$ were ± 1 then every polynomial would be a multiple of $a(x)$, contrary to $(a(x))$ being a proper ideal. The only possibility is $a(x) = \pm 2$. But now $x \in (a(x)) = (2) = (-2)$ and so $x = 2q(x)$ for some polynomial $q(x)$ with integer coefficients, clearly impossible. This contradiction proves that $(2, x)$ is not principal.

Note that the symbol (A) is ambiguous if the ring is not specified: the ideal generated by 2 and x in $\mathbb{Q}[x]$ is the entire ring (1) since it contains the element $\frac{1}{2}2 = 1$.

We shall see in Chapter 9 that for any *field* F , all ideals of $F[x]$ are principal.

- (4) If R is the ring of all functions from the closed interval $[0, 1]$ into $\mathbb{R}$ let M be the ideal $\{f \mid f(\frac{1}{2}) = 0\}$ (the kernel of evaluation at $\frac{1}{2}$). Let $g(x)$ be the function which is zero at $x = \frac{1}{2}$ and 1 at all other points. Then $f = fg$ for all $f \in M$ so M is a principal ideal with generator g . In fact, *any* function which is zero at $\frac{1}{2}$ and nonzero at all other points is another generator for the same ideal M .

On the other hand, if R is the ring of all *continuous* functions from $[0, 1]$ to $\mathbb{R}$ then $\{f \mid f(\frac{1}{2}) = 0\}$ is *not* principal nor is it even finitely generated (cf. the exercises).

- (5) If G is a finite group and R is a commutative ring with 1 then the augmentation ideal is generated by the set $\{g - 1 \mid g \in G\}$, although this need not be a minimal set of generators. For example, if G is a cyclic group with generator σ , then the augmentation ideal is a principal ideal with generator $\sigma - 1$.

Proposition 9. Let I be an ideal of R .

- (1) $I = R$ if and only if I contains a unit.
 (2) Assume R is commutative. Then R is a field if and only if its only ideals are 0 and R .

Proof: (1) If $I = R$ then I contains the unit 1. Conversely, if u is a unit in I with inverse v , then for any $r \in R$

$$r = r \cdot 1 = r(vu) = (rv)u \in I$$

hence $R = I$.

(2) The ring R is a field if and only if every nonzero element is a unit. If R is a field every nonzero ideal contains a unit, so by the first part R is the only nonzero ideal. Conversely, if 0 and R are the only ideals of R let u be any nonzero element of R . By hypothesis $(u) = R$ and so $1 \in (u)$. Thus there is some $v \in R$ such that $1 = vu$, i.e., u is a unit. Every nonzero element of R is therefore a unit and so R is a field.

Corollary 10. If R is a field then any nonzero ring homomorphism from R into another ring is an injection.

Proof: The kernel of a ring homomorphism is an ideal. The kernel of a nonzero homomorphism is a proper ideal hence is 0 by the proposition.

These results show that the ideal structure of fields is trivial. Our approach to studying an algebraic structure through its homomorphisms will still play a fundamental role in field theory (Part IV) when we study injective homomorphisms (embeddings) of one field into another and automorphisms of fields (isomorphisms of a field to itself).

If D is a ring with identity $1 \neq 0$ in which the only left ideals and the only right ideals are 0 and D , then D is a division ring. Conversely, the only (left, right or two-sided) ideals in a division ring D are 0 and D , which gives an analogue of Proposition 9(2) if R is not commutative (see the exercises). However, if F is a field, then for any $n \geq 2$ the only two-sided ideals in the matrix ring $M_n(F)$ are 0 and $M_n(F)$, even though this is not a division ring (it does have proper, nontrivial, left and right ideals: cf. Section 3), which shows that Proposition 9(2) does not hold for noncommutative rings. Rings whose only two-sided ideals are 0 and the whole ring (which are called *simple rings*) will be studied in Chapter 18.

One important class of ideals are those which are not contained in any other proper ideal:

Definition. An ideal M in an arbitrary ring S is called a *maximal ideal* if $M \neq S$ and the only ideals containing M are M and S .

A general ring need not have maximal ideals. For example, take any abelian group which has no maximal subgroups (for example, $\mathbb{Q}$ — cf. Exercise 16, Section 6.1) and make it into a trivial ring by defining $ab = 0$ for all a, b . In such a ring the ideals are simply the subgroups and so there are no maximal ideals. The zero ring has no maximal ideals, hence any result involving maximal ideals forces a ring to be nonzero. The next proposition shows that rings with an identity $1 \neq 0$ always possess maximal ideals. Like many such general existence theorems (e.g., the result that a finitely generated group has maximal subgroups or that every vector space has a basis) the proof relies on Zorn's Lemma (see Appendix I). In many specific rings, however, the presence of maximal ideals is often obvious, independent of Zorn's Lemma.

Proposition 11. In a ring with identity every proper ideal is contained in a maximal ideal.

Proof: Let R be a ring with identity and let I be a proper ideal (so R cannot be the zero ring, i.e., $1 \neq 0$). Let S be the set of all proper ideals of R which contain I . Then S is nonempty ($I \in S$) and is partially ordered by inclusion. If C is a chain in S , define J to be the union of all ideals in C :

$$J = \bigcup_{A \in C} A.$$

We first show that J is an ideal. Certainly J is nonempty because C is nonempty — specifically, $0 \in J$ since 0 is in every ideal A . If $a, b \in J$, then there are ideals $A, B \in C$ such that $a \in A$ and $b \in B$. By definition of a chain either $A \subseteq B$ or $B \subseteq A$. In either case $a - b \in J$, so J is closed under subtraction. Since each $A \in C$ is closed under left and right multiplication by elements of R , so is J . This proves J is an ideal.

If J is not a proper ideal then $1 \in J$. In this case, by definition of J we must have $1 \in A$ for some $A \in C$. This is a contradiction because each A is a proper ideal ($A \in C \subseteq S$). This proves that each chain has an upper bound in S . By Zorn's Lemma S has a maximal element which is therefore a maximal (proper) ideal containing I .

For commutative rings the next result characterizes maximal ideals by the structure of their quotient rings.

Proposition 12. Assume R is commutative. The ideal M is a maximal ideal if and only if the quotient ring R/M is a field.

Proof: This follows from the Lattice Isomorphism Theorem together with Proposition 9(2). The ideal M is maximal if and only if there are no ideals I with $M \subset I \subset R$. By the Lattice Isomorphism Theorem the ideals of R containing M correspond bijectively with the ideals of R/M , so M is maximal if and only if the only ideals of R/M are 0 and R/M . By Proposition 9(2) we see that M is maximal if and only if R/M is a field.

The proposition above indicates how to *construct* some fields: take the quotient of any commutative ring R with identity by a maximal ideal in R . We shall use this in Part IV to construct all finite fields by taking quotients of the ring $\mathbb{Z}[x]$ by maximal ideals.

Examples

- (1) Let n be a nonnegative integer. The ideal $n\mathbb{Z}$ of $\mathbb{Z}$ is a maximal ideal if and only if $\mathbb{Z}/n\mathbb{Z}$ is a field. We saw in Section 3 that this is the case if and only if n is a prime number. This also follows directly from the containment of ideals of $\mathbb{Z}$ described in Example 2 above.
- (2) The ideal $(2, x)$ is a maximal ideal in $\mathbb{Z}[x]$ because its quotient ring is the field $\mathbb{Z}/2\mathbb{Z}$ — cf. Example 3 above and Example 5 at the end of Section 3.
- (3) The ideal (x) in $\mathbb{Z}[x]$ is not a maximal ideal because $(x) \subset (2, x) \subset \mathbb{Z}[x]$. The quotient ring $\mathbb{Z}[x]/(x)$ is isomorphic to $\mathbb{Z}$ (the ideal (x) in $\mathbb{Z}[x]$ is the kernel of the surjective ring homomorphism from $\mathbb{Z}[x]$ to $\mathbb{Z}$ given by evaluation at 0). Since $\mathbb{Z}$ is not a field, we see again that (x) is not a maximal ideal in $\mathbb{Z}[x]$.
- (4) Let R be the ring of all functions from $[0, 1]$ to $\mathbb{R}$ and for each $a \in [0, 1]$ let M_a be the kernel of evaluation at a . Since evaluation is a surjective homomorphism from R to $\mathbb{R}$, we see that $R/M_a \cong \mathbb{R}$ and hence M_a is a maximal ideal. Similarly, the kernel of evaluation at any fixed point is a maximal ideal in the ring of continuous real valued functions on $[0, 1]$.
- (5) If F is a field and G is a finite group, then the augmentation ideal I is a maximal ideal of the group ring FG (cf. Example 7 at the end of the preceding section). The augmentation ideal is the kernel of the augmentation map which is a surjective homomorphism onto the field F (i.e., $FG/I \cong F$, a field). Note that Proposition 12 does not apply directly since FG need not be commutative, however, the implication in Proposition 12 that I is a maximal ideal if R/I is a field holds for arbitrary rings.

Definition. Assume R is commutative. An ideal P is called a *prime ideal* if $P \neq R$ and whenever the product ab of two elements $a, b \in R$ is an element of P , then at least one of a and b is an element of P .

The notion of a maximal ideal is fairly intuitive but the definition of a prime ideal may seem a little strange. It is, however, a natural generalization of the notion of a “prime” in the integers $\mathbb{Z}$. Let n be a nonnegative integer. According to the above definition the ideal $n\mathbb{Z}$ is a *prime ideal* provided $n \neq 1$ (to ensure that the ideal is proper) and provided every time the product ab of two integers is an element of $n\mathbb{Z}$, at least one of a, b is an element of $n\mathbb{Z}$. Put another way, if $n \neq 0$, it must have the property that whenever n divides ab , n must divide a or divide b . This is equivalent to the usual definition that n is a prime number. Thus *the prime ideals of $\mathbb{Z}$ are just the ideals $p\mathbb{Z}$ of $\mathbb{Z}$ generated by prime numbers p together with the ideal 0 .*

For the integers $\mathbb{Z}$ there is no difference between the maximal ideals and the nonzero prime ideals. This is not true in general, but we shall see shortly that every maximal ideal is a prime ideal. First we translate the notion of prime ideals into properties of quotient rings as we did for maximal ideals in Proposition 12. Recall that an integral domain is a commutative ring with identity $1 \neq 0$ that has no zero divisors.

Proposition 13. Assume R is commutative. Then the ideal P is a prime ideal in R if and only if the quotient ring R/P is an integral domain.

Proof: This proof is simply a matter of translating the definition of a prime ideal into the language of quotients. The ideal P is prime if and only if $P \neq R$ and whenever

$ab \in P$, then either $a \in P$ or $b \in P$. Use the bar notation for elements of R/P : $\bar{r} = r + P$. Note that $r \in P$ if and only if the element $\bar{r}$ is zero in the quotient ring R/P . Thus in the terminology of quotients P is a prime ideal if and only if $\bar{R} \neq \bar{0}$ and whenever $\overline{ab} = \bar{a}\bar{b} = \bar{0}$, then either $\bar{a} = \bar{0}$ or $\bar{b} = \bar{0}$, i.e., R/P is an integral domain.

It follows in particular that a commutative ring with identity is an integral domain if and only if 0 is a prime ideal.

Corollary 14. Assume R is commutative. Every maximal ideal of R is a prime ideal.

Proof: If M is a maximal ideal then R/M is a field by Proposition 12. A field is an integral domain so the corollary follows from Proposition 13.

Examples

- (1) The principal ideals generated by primes in $\mathbb{Z}$ are both prime and maximal ideals. The zero ideal in $\mathbb{Z}$ is prime but not maximal.
- (2) The ideal (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. This ideal is not a maximal ideal. The ideal 0 is a prime ideal in $\mathbb{Z}[x]$, but is not a maximal ideal.

EXERCISES

Let R be a ring with identity $1 \neq 0$.

1. Let L_j be the left ideal of $M_n(R)$ consisting of arbitrary entries in the j^{th} column and zero in all other entries and let E_{ij} be the element of $M_n(R)$ whose i, j entry is 1 and whose other entries are all 0. Prove that $L_j = M_n(R)E_{ij}$ for any i . [See Exercise 6, Section 2.]
2. Assume R is commutative. Prove that the augmentation ideal in the group ring RG is generated by $\{g - 1 \mid g \in G\}$. Prove that if $G = \langle \sigma \rangle$ is cyclic then the augmentation ideal is generated by $\sigma - 1$.
3. (a) Let p be a prime and let G be an abelian group of order p^n . Prove that the nilradical of the group ring $\mathbb{F}_p G$ is the augmentation ideal (cf. Exercise 29, Section 3). [Use the preceding exercise.]
 (b) Let $G = \{g_1, \dots, g_n\}$ be a finite group and assume R is commutative. Prove that if r is any element of the augmentation ideal of RG then $r(g_1 + \dots + g_n) = 0$. [Use the preceding exercise.]
4. Assume R is commutative. Prove that R is a field if and only if 0 is a maximal ideal.
5. Prove that if M is an ideal such that R/M is a field then M is a maximal ideal (do not assume R is commutative).
6. Prove that R is a division ring if and only if its only left ideals are (0) and R . (The analogous result holds when “left” is replaced by “right.”)
7. Let R be a commutative ring with 1. Prove that the principal ideal generated by x in the polynomial ring $R[x]$ is a prime ideal if and only if R is an integral domain. Prove that (x) is a maximal ideal if and only if R is a field.
8. Let R be an integral domain. Prove that $(a) = (b)$ for some elements $a, b \in R$, if and only if $a = ub$ for some unit u of R .
9. Let R be the ring of all continuous functions on $[0, 1]$ and let I be the collection of functions $f(x)$ in R with $f(1/3) = f(1/2) = 0$. Prove that I is an ideal of R but is not a prime ideal.

10. Assume R is commutative. Prove that if P is a prime ideal of R and P contains no zero divisors then R is an integral domain.
11. Assume R is commutative. Let I and J be ideals of R and assume P is a prime ideal of R that contains IJ (for example, if P contains $I \cap J$). Prove either I or J is contained in P .
12. Assume R is commutative and suppose $I = (a_1, a_2, \dots, a_n)$ and $J = (b_1, b_2, \dots, b_m)$ are two finitely generated ideals in R . Prove that the product ideal IJ is finitely generated by the elements $a_i b_j$ for $i = 1, 2, \dots, n$, and $j = 1, 2, \dots, m$.
13. Let $\varphi : R \rightarrow S$ be a homomorphism of commutative rings.
- (a) Prove that if P is a prime ideal of S then either $\varphi^{-1}(P) = R$ or $\varphi^{-1}(P)$ is a prime ideal of R . Apply this to the special case when R is a subring of S and φ is the inclusion homomorphism to deduce that if P is a prime ideal of S then $P \cap R$ is either R or a prime ideal of R .
- (b) Prove that if M is a maximal ideal of S and φ is surjective then $\varphi^{-1}(M)$ is a maximal ideal of R . Give an example to show that this need not be the case if φ is not surjective.
14. Assume R is commutative. Let x be an indeterminate, let $f(x)$ be a monic polynomial in $R[x]$ of degree $n \geq 1$ and use the bar notation to denote passage to the quotient ring $R[x]/(f(x))$.

- (a) Show that every element of $R[x]/(f(x))$ is of the form $\overline{p(x)}$ for some polynomial $p(x) \in R[x]$ of degree less than n , i.e.,

$$R[x]/(f(x)) = \{\overline{a_0} + \overline{a_1}x + \cdots + \overline{a_{n-1}}x^{n-1} \mid a_0, a_1, \dots, a_{n-1} \in R\}.$$

[If $f(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_0$ then $\overline{x^n} = -\overline{(b_{n-1}x^{n-1} + \cdots + b_0)}$. Use this to reduce powers of $\overline{x}$ in the quotient ring.]

- (b) Prove that if $\overline{p(x)}$ and $\overline{q(x)}$ are distinct polynomials in $R[x]$ which are both of degree less than n , then $\overline{p(x)} \neq \overline{q(x)}$. [Otherwise $p(x) - q(x)$ is an $R[x]$ -multiple of the monic polynomial $f(x)$.]
- (c) If $f(x) = a(x)b(x)$ where both $a(x)$ and $b(x)$ have degree less than n , prove that $\overline{a(x)}$ is a zero divisor in $R[x]/(f(x))$.
- (d) If $f(x) = x^n - a$ for some nilpotent element $a \in R$, prove that $\overline{x}$ is nilpotent in $R[x]/(f(x))$.
- (e) Let p be a prime, assume $R = \mathbb{F}_p$ and $f(x) = x^p - a$ for some $a \in \mathbb{F}_p$. Prove that $\overline{x - a}$ is nilpotent in $R[x]/(f(x))$. [Use Exercise 26(c) of Section 3.]
15. Let $x^2 + x + 1$ be an element of the polynomial ring $E = \mathbb{F}_2[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{F}_2[x]/(x^2 + x + 1)$.
- (a) Prove that $\overline{E}$ has 4 elements: $\overline{0}$, $\overline{1}$, $\overline{x}$ and $\overline{x+1}$.
- (b) Write out the 4×4 addition table for $\overline{E}$ and deduce that the additive group $\overline{E}$ is isomorphic to the Klein 4-group.
- (c) Write out the 4×4 multiplication table for $\overline{E}$ and prove that $\overline{E}^\times$ is isomorphic to the cyclic group of order 3. Deduce that $\overline{E}$ is a field.
16. Let $x^4 - 16$ be an element of the polynomial ring $E = \mathbb{Z}[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{Z}[x]/(x^4 - 16)$.
- (a) Find a polynomial of degree ≤ 3 that is congruent to $7x^{13} - 11x^9 + 5x^5 - 2x^3 + 3$ modulo $(x^4 - 16)$.
- (b) Prove that $\overline{x - 2}$ and $\overline{x + 2}$ are zero divisors in $\overline{E}$.
17. Let $x^3 - 2x + 1$ be an element of the polynomial ring $E = \mathbb{Z}[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{Z}[x]/(x^3 - 2x + 1)$. Let $p(x) = 2x^7 - 7x^5 + 4x^3 - 9x + 1$ and let $q(x) = (x - 1)^4$.

- (a) Express each of the following elements of $\overline{E}$ in the form $\overline{f(x)}$ for some polynomial $f(x)$ of degree ≤ 2 : $\overline{p(x)}$, $\overline{q(x)}$, $\overline{p(x) + q(x)}$ and $\overline{p(x)q(x)}$.
- (b) Prove that $\overline{E}$ is not an integral domain.
- (c) Prove that $\overline{x}$ is a unit in $\overline{E}$.
18. Prove that if R is an integral domain and $R[[x]]$ is the ring of formal power series in the indeterminate x then the principal ideal generated by x is a prime ideal (cf. Exercise 3, Section 2). Prove that the principal ideal generated by x is a maximal ideal if and only if R is a field.
19. Let R be a finite commutative ring with identity. Prove that every prime ideal of R is a maximal ideal.
20. Prove that a nonzero finite commutative ring that has no zero divisors is a field (if the ring has an identity, this is Corollary 3, so do not assume the ring has a 1).
21. Prove that a finite ring with identity $1 \neq 0$ that has no zero divisors is a field (you may quote Wedderburn's Theorem).
22. Let $p \in \mathbb{Z}^+$ be a prime and let the $\mathbb{F}_p$ Quaternions be defined by

$$a + bi + cj + dk \quad a, b, c, d \in \mathbb{Z}/p\mathbb{Z}$$

where addition is componentwise and multiplication is defined using the same relations on i, j, k as for the real Quaternions.

- (a) Prove that the $\mathbb{F}_p$ Quaternions are a homomorphic image of the integral Quaternions (cf. Section 1).
- (b) Prove that the $\mathbb{F}_p$ Quaternions contain zero divisors (and so they cannot be a division ring). [Use the preceding exercise.]
23. Prove that in a Boolean ring (cf. Exercise 15, Section 1) every prime ideal is a maximal ideal.
24. Prove that in a Boolean ring every finitely generated ideal is principal.
25. Assume R is commutative and for each $a \in R$ there is an integer $n > 1$ (depending on a) such that $a^n = a$. Prove that every prime ideal of R is a maximal ideal.
26. Prove that a prime ideal in a commutative ring R contains every nilpotent element (cf. Exercise 13, Section 1). Deduce that the nilradical of R (cf. Exercise 29, Section 3) is contained in the intersection of all the prime ideals of R . (It is shown in Section 15.2 that the nilradical of R is equal to the intersection of all prime ideals of R .)
27. Let R be a commutative ring with $1 \neq 0$. Prove that if a is a nilpotent element of R then $1 - ab$ is a unit for all $b \in R$.
28. Prove that if R is a commutative ring and $N = (a_1, a_2, \dots, a_m)$ where each a_i is a nilpotent element, then N is a nilpotent ideal (cf. Exercise 37, Section 3). Deduce that if the nilradical of R is finitely generated then it is a nilpotent ideal.
29. Let p be a prime and let G be a finite group of order a power of p (i.e., a p -group). Prove that the augmentation ideal in the group ring $\mathbb{Z}/p\mathbb{Z}G$ is a nilpotent ideal. (Note that this ring may be noncommutative.) [Use Exercise 2.]
30. Let I be an ideal of the commutative ring R and define

$$\text{rad } I = \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{Z}^+\}$$

called the *radical* of I . Prove that $\text{rad } I$ is an ideal containing I and that $(\text{rad } I)/I$ is the nilradical of the quotient ring R/I , i.e., $(\text{rad } I)/I = \mathfrak{N}(R/I)$ (cf. Exercise 29, Section 3).

31. An ideal I of the commutative ring R is called a *radical ideal* if $\text{rad } I = I$.

- (a) Prove that every prime ideal of R is a radical ideal.
- (b) Let $n > 1$ be an integer. Prove that 0 is a radical ideal in $\mathbb{Z}/n\mathbb{Z}$ if and only if n is a product of distinct primes to the first power (i.e., n is square free). Deduce that (n) is a radical ideal of $\mathbb{Z}$ if and only if n is a product of distinct primes in $\mathbb{Z}$.
32. Let I be an ideal of the commutative ring R and define
- $$\text{Jac } I \text{ to be the intersection of all maximal ideals of } R \text{ that contain } I$$
- where the convention is that $\text{Jac } R = R$. (If I is the zero ideal, $\text{Jac } 0$ is called the *Jacobson radical* of the ring R , so $\text{Jac } I$ is the preimage in R of the Jacobson radical of R/I .)
- (a) Prove that $\text{Jac } I$ is an ideal of R containing I .
- (b) Prove that $\text{rad } I \subseteq \text{Jac } I$, where $\text{rad } I$ is the radical of I defined in Exercise 30.
- (c) Let $n > 1$ be an integer. Describe $\text{Jac } n\mathbb{Z}$ in terms of the prime factorization of n .
33. Let R be the ring of all continuous functions from the closed interval $[0,1]$ to $\mathbb{R}$ and for each $c \in [0, 1]$ let $M_c = \{f \in R \mid f(c) = 0\}$ (recall that M_c was shown to be a maximal ideal of R).
- (a) Prove that if M is any maximal ideal of R then there is a real number $c \in [0, 1]$ such that $M = M_c$.
- (b) Prove that if b and c are distinct points in $[0,1]$ then $M_b \neq M_c$.
- (c) Prove that M_c is not equal to the principal ideal generated by $x - c$.
- (d) Prove that M_c is not a finitely generated ideal.

The preceding exercise shows that there is a bijection between the *points* of the closed interval $[0,1]$ and the set of *maximal ideals* in the ring R of all of continuous functions on $[0,1]$ given by $c \leftrightarrow M_c$. For any subset X of $\mathbb{R}$ or, more generally, for any completely regular topological space X , the map $c \mapsto M_c$ is an *injection* from X to the set of maximal ideals of R , where R is the ring of all bounded continuous real valued functions on X and M_c is the maximal ideal of functions that vanish at c . Let $\beta(X)$ be the set of maximal ideals of R . One can put a topology on $\beta(X)$ in such a way that if we identify X with its image in $\beta(X)$ then X (in its given topology) becomes a subspace of $\beta(X)$. Moreover, $\beta(X)$ is a compact space under this topology and is called the *Stone-Čech compactification* of X .

34. Let R be the ring of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ and for each $c \in \mathbb{R}$ let M_c be the maximal ideal $\{f \in R \mid f(c) = 0\}$.
- (a) Let I be the collection of functions $f(x)$ in R with *compact support* (i.e., $f(x) = 0$ for $|x|$ sufficiently large). Prove that I is an ideal of R that is not a prime ideal.
- (b) Let M be a maximal ideal of R containing I (properly, by (a)). Prove that $M \neq M_c$ for any $c \in \mathbb{R}$ (cf. the preceding exercise).
35. Let $A = (a_1, a_2, \dots, a_n)$ be a nonzero finitely generated ideal of R . Prove that there is an ideal B which is maximal with respect to the property that it does not contain A . [Use Zorn's Lemma.]
36. Assume R is commutative. Prove that the set of prime ideals in R has a minimal element with respect to inclusion (possibly the zero ideal). [Use Zorn's Lemma.]
37. A commutative ring R is called a *local ring* if it has a unique maximal ideal. Prove that if R is a local ring with maximal ideal M then every element of $R - M$ is a unit. Prove conversely that if R is a commutative ring with 1 in which the set of nonunits forms an ideal M , then R is a local ring with unique maximal ideal M .
38. Prove that the ring of all rational numbers whose denominators is odd is a local ring whose unique maximal ideal is the principal ideal generated by 2 .
39. Following the notation of Exercise 26 in Section 1, let K be a field, let v be a discrete

valuation on K and let R be the valuation ring of v . For each integer $k \geq 0$ define $A_k = \{r \in R \mid v(r) \geq k\} \cup \{0\}$.

(a) Prove that A_k is a principal ideal and that $A_0 \supseteq A_1 \supseteq A_2 \supseteq \dots$.

(b) Prove that if I is any nonzero ideal of R , then $I = A_k$ for some $k \geq 0$. Deduce that R is a local ring with unique maximal ideal A_1 .

40. Assume R is commutative. Prove that the following are equivalent: (see also Exercises 13 and 14 in Section 1)

(i) R has exactly one prime ideal

(ii) every element of R is either nilpotent or a unit

(iii) $R/\eta(R)$ is a field (cf. Exercise 29, Section 3).

41. A proper ideal Q of the commutative ring R is called *primary* if whenever $ab \in Q$ and $a \notin Q$ then $b^n \in Q$ for some positive integer n . (Note that the symmetry between a and b in this definition implies that if Q is a primary ideal and $ab \in Q$ with *neither* a nor b in Q , then a positive power of a and a positive power of b both lie in Q .) Establish the following facts about primary ideals.

(a) The primary ideals of $\mathbb{Z}$ are 0 and (p^n) , where p is a prime and n is a positive integer.

(b) Every prime ideal of R is a primary ideal.

(c) An ideal Q of R is primary if and only if every zero divisor in R/Q is a nilpotent element of R/Q .

(d) If Q is a primary ideal then $\text{rad}(Q)$ is a prime ideal (cf. Exercise 30).

7.5 RINGS OF FRACTIONS

Throughout this section R is a commutative ring. Proposition 2 shows that if a is not zero nor a zero divisor and $ab = ac$ in R then $b = c$. Thus a nonzero element that is not a zero divisor enjoys some of the properties of a unit without necessarily possessing a multiplicative inverse in R . On the other hand, we saw in Section 1 that a zero divisor a cannot be a unit in R and, by definition, if a is a zero divisor we cannot always cancel the a 's in the equation $ab = ac$ to obtain $b = c$ (take $c = 0$ for example). The aim of this section is to prove that a commutative ring R is always a subring of a larger ring Q in which every nonzero element of R that is not a zero divisor is a unit in Q . The principal application of this will be to integral domains, in which case this ring Q will be a field — called its *field of fractions* or *quotient field*. Indeed, the paradigm for the construction of Q from R is the one offered by the construction of the field of rational numbers from the integral domain $\mathbb{Z}$.

In order to see the essential features of the construction of the field $\mathbb{Q}$ from the integral domain $\mathbb{Z}$ we review the basic properties of fractions. Each rational number may be represented in many different ways as the quotient of two integers (for example,

$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \dots$, etc.). These representations are related by

$$\frac{a}{b} = \frac{c}{d} \quad \text{if and only if} \quad ad = bc.$$

In more precise terms, the fraction $\frac{a}{b}$ is the equivalence class of ordered pairs (a, b) of integers with $b \neq 0$ under the equivalence relation: $(a, b) \sim (c, d)$ if and only if

$ad = bc$. The arithmetic operations on fractions are given by

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

These are well defined (independent of choice of representatives of the equivalence classes) and make the set of fractions into a commutative ring (in fact, a field), $\mathbb{Q}$. The integers $\mathbb{Z}$ are identified with the subring $\{\frac{a}{1} \mid a \in \mathbb{Z}\}$ of $\mathbb{Q}$ and every nonzero integer a has an inverse $\frac{1}{a}$ in $\mathbb{Q}$.

It seems reasonable to attempt to follow the same steps for any commutative ring R , allowing arbitrary denominators. If, however, b is zero or a zero divisor in R , say $bd = 0$, and if we allow b as a denominator, then we should expect to have

$$d = \frac{d}{1} = \frac{bd}{b} = \frac{0}{b} = 0$$

in the “ring of fractions” (where, for convenience, we have assumed R has a 1). Thus if we allow zero or zero divisors as denominators there must be some collapsing in the sense that we cannot expect R to appear naturally as a subring of this “ring of fractions.” A second restriction is more obviously imposed by the laws of addition and multiplication: if ring elements b and d are allowed as denominators, then bd must also be a denominator, i.e., the set of denominators must be closed under multiplication in R . The main result of this section shows that these two restrictions are sufficient to construct a ring of fractions for R . Note that this theorem includes the construction of $\mathbb{Q}$ from $\mathbb{Z}$ as a special case.

Theorem 15. Let R be a commutative ring. Let D be any nonempty subset of R that does not contain 0, does not contain any zero divisors and is closed under multiplication (i.e., $ab \in D$ for all $a, b \in D$). Then there is a commutative ring Q with 1 such that Q contains R as a subring and every element of D is a unit in Q . The ring Q has the following additional properties.

- (1) every element of Q is of the form rd^{-1} for some $r \in R$ and $d \in D$. In particular, if $D = R - \{0\}$ then Q is a field.
- (2) (uniqueness of Q) The ring Q is the “smallest” ring containing R in which all elements of D become units, in the following sense. Let S be any commutative ring with identity and let $\varphi : R \rightarrow S$ be any injective ring homomorphism such that $\varphi(d)$ is a unit in S for every $d \in D$. Then there is an injective homomorphism $\Phi : Q \rightarrow S$ such that $\Phi|_R = \varphi$. In other words, any ring containing an isomorphic copy of R in which all the elements of D become units must also contain an isomorphic copy of Q .

Remark: In Section 15.4 a more general construction is given. The proof of the general result is more technical but relies on the same basic rationale and steps as the proof of Theorem 15. Readers wishing greater generality may read the proof below and the beginning of Section 15.4 in concert.

Proof: Let $\mathcal{F} = \{(r, d) \mid r \in R, d \in D\}$ and define the relation $\sim$ on $\mathcal{F}$ by

$$(r, d) \sim (s, e) \quad \text{if and only if} \quad re = sd.$$

It is immediate that this relation is reflexive and symmetric. Suppose $(r, d) \sim (s, e)$ and $(s, e) \sim (t, f)$. Then $re - sd = 0$ and $sf - te = 0$. Multiplying the first of these equations by f and the second by d and adding them gives $(rf - td)e = 0$. Since $e \in D$ is neither zero nor a zero divisor we must have $rf - td = 0$, i.e., $(r, d) \sim (t, f)$. This proves $\sim$ is transitive, hence an equivalence relation. Denote the equivalence class of (r, d) by $\frac{r}{d}$:

$$\frac{r}{d} = \{(a, b) \mid a \in R, b \in D \text{ and } rb = ad\}.$$

Let Q be the set of equivalence classes under $\sim$. Note that $\frac{r}{d} = \frac{re}{de}$ in Q for all $e \in D$, since D is closed under multiplication.

We now define an additive and multiplicative structure on Q :

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}.$$

In order to prove that Q is a commutative ring with identity there are a number of things to check:

- (1) these operations are well defined (i.e., do not depend on the choice of representatives for the equivalence classes),
- (2) Q is an abelian group under addition, where the additive identity is $\frac{0}{d}$ for any $d \in D$ and the additive inverse of $\frac{a}{d}$ is $\frac{-a}{d}$,
- (3) multiplication is associative, distributive and commutative, and
- (4) Q has an identity ($= \frac{d}{d}$ for any $d \in D$).

These are all completely straightforward calculations involving only arithmetic in R and the definition of $\sim$. Again we need D to be closed under multiplication for addition and multiplication to be defined.

For example, to check that addition is well defined assume $\frac{a}{b} = \frac{a'}{b'}$ (i.e., $ab' = a'b$) and $\frac{c}{d} = \frac{c'}{d'}$ (i.e., $cd' = c'd$). We must show that $\frac{ad + bc}{bd} = \frac{a'd' + b'c'}{b'd'}$, i.e.,

$$(ad + bc)(b'd') = (a'd' + b'c')(bd).$$

The left hand side of this equation is $ab'dd' + cd'bb'$ substituting $a'b$ for ab' and $c'd$ for cd' gives $a'bdd' + c'dbb'$, which is the right hand side. Hence addition of fractions is well defined. Checking the details in the other parts of (1) to (4) involves even easier manipulations and so is left as an exercise.

Next we embed R into Q by defining

$$\iota : R \rightarrow Q \quad \text{by} \quad \iota : r \mapsto \frac{rd}{d} \quad \text{where } d \text{ is any element of } D.$$

Since $\frac{rd}{d} = \frac{re}{e}$ for all $d, e \in D$, $\iota(r)$ does not depend on the choice of $d \in D$. Since D is closed under multiplication, one checks directly that ι is a ring homomorphism.

Furthermore, ι is injective because

$$\iota(r) = 0 \Leftrightarrow \frac{rd}{d} = \frac{0}{d} \Leftrightarrow rd^2 = 0 \Leftrightarrow r = 0$$

because d (hence also d^2) is neither zero nor a zero divisor. The subring $\iota(R)$ of Q is therefore isomorphic to R . We henceforth identify each $r \in R$ with $\iota(r)$ and so consider R as a subring of Q .

Next note that each $d \in D$ has a multiplicative inverse in Q : namely, if d is represented by the fraction $\frac{de}{e}$ then its multiplicative inverse is $\frac{e}{de}$. One then sees that every element of Q may be written as $r \cdot d^{-1}$ for some $r \in R$ and some $d \in D$. In particular, if $D = R - \{0\}$, every nonzero element of Q has a multiplicative inverse and Q is a field.

It remains to establish the uniqueness property of Q . Assume $\varphi : R \rightarrow S$ is an injective ring homomorphism such that $\varphi(d)$ is a unit in S for all $d \in D$. Extend φ to a map $\Phi : Q \rightarrow S$ by defining $\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1}$ for all $r \in R, d \in D$. This map is well defined, since $rd^{-1} = se^{-1}$ implies $re = sd$, so $\varphi(r)\varphi(e) = \varphi(s)\varphi(d)$, and then

$$\Phi(rd^{-1}) = \varphi(r)\varphi(d)^{-1} = \varphi(s)\varphi(e)^{-1} = \Phi(se^{-1}).$$

It is straightforward to check that Φ is a ring homomorphism — the details are left as an exercise. Finally, Φ is injective because $rd^{-1} \in \ker \Phi$ implies $r \in \ker \Phi \cap R = \ker \varphi$; since φ is injective this forces r and hence also rd^{-1} to be zero. This completes the proof.

Definition. Let R, D and Q be as in Theorem 15.

- (1) The ring Q is called the *ring of fractions* of D with respect to R and is denoted $D^{-1}R$.
- (2) If R is an integral domain and $D = R - \{0\}$, Q is called the *field of fractions* or *quotient field* of R .

If A is a subset of a field F (for example, if A is a subring of F), then the intersection of all the subfields of F containing A is a subfield of F and is called the subfield *generated* by A . This subfield is the smallest subfield of F containing A (namely, any subfield of F containing A contains the subfield generated by A).

The next corollary shows that the smallest field containing an integral domain R is its field of fractions.

Corollary 16. Let R be an integral domain and let Q be the field of fractions of R . If a field F contains a subring R' isomorphic to R then the subfield of F generated by R' is isomorphic to Q .

Proof: Let $\varphi : R \cong R' \subseteq F$ be a (ring) isomorphism of R to R' . In particular, $\varphi : R \rightarrow F$ is an injective homomorphism from R into the field F . Let $\Phi : Q \rightarrow F$ be the extension of φ to Q as in the theorem. By Theorem 15, Φ is injective, so $\Phi(Q)$ is an isomorphic copy of Q in F containing $\varphi(R) = R'$. Now, any subfield of F containing $R' = \varphi(R)$ contains the elements $\varphi(r_1)\varphi(r_2)^{-1} = \varphi(r_1r_2^{-1})$ for all $r_1, r_2 \in R$. Since

every element of Q is of the form $r_1 r_2^{-1}$ for some $r_1, r_2 \in R$, it follows that any subfield of F containing R' contains the field $\Phi(Q)$, so that $\Phi(Q)$ is the subfield of F generated by R' , proving the corollary.

Examples

- (1) If R is a field then its field of fractions is just R itself.
- (2) The integers $\mathbb{Z}$ are an integral domain whose field of fractions is the field $\mathbb{Q}$ of rational numbers. The quadratic integer ring $\mathcal{O}$ of Section 1 is an integral domain whose field of fractions is the quadratic field $\mathbb{Q}(\sqrt{D})$.
- (3) The subring $2\mathbb{Z}$ of $\mathbb{Z}$ also has no zero divisors (but has no identity). Its field of fractions is also $\mathbb{Q}$. Note how an identity “appears” in the field of fractions.
- (4) If R is any integral domain, then the polynomial ring $R[x]$ is also an integral domain. The associated field of fractions is the field of *rational functions* in the variable x over R . The elements of this field are of the form $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials with coefficients in R with $q(x)$ not the zero polynomial. In particular, $p(x)$ and $q(x)$ may both be constant polynomials, so the field of rational functions contains the field of fractions of R : elements of the form $\frac{a}{b}$ such that $a, b \in R$ and $b \neq 0$. If F is a field, we shall denote the field of rational functions by $F(x)$. Thus if F is the field of fractions of the integral domain R then the field of rational functions over R is the same as the field of rational functions over F , namely $F(x)$.

For example, suppose $R = \mathbb{Z}$, so $F = \mathbb{Q}$. If $p(x), q(x)$ are polynomials in $\mathbb{Q}[x]$ then for some integer N , $Np(x), Nq(x)$ have integer coefficients (let N be a common denominator for all the coefficients in $p(x)$ and $q(x)$, for example). Then $\frac{p(x)}{q(x)} = \frac{Np(x)}{Nq(x)}$ can be written as the quotient of two polynomials with integer coefficients, so the field of fractions of $\mathbb{Q}[x]$ is the same as the field of fractions of $\mathbb{Z}[x]$.

- (5) If R is any commutative ring with identity and d is neither zero nor a zero divisor in R we may form the ring $R[1/d]$ by setting $D = \{1, d, d^2, d^3, \dots\}$ and defining $R[1/d]$ to be the ring of fractions $D^{-1}R$. Note that R is the subring of elements of the form $\frac{r}{1}$. In this way any nonzero element of R that is not a zero divisor can be inverted in a larger ring containing R . Note that the elements of $R[1/d]$ look like polynomials in $1/d$ with coefficients in R , which explains the notation.

EXERCISES

Let R be a commutative ring with identity $1 \neq 0$.

1. Fill in all the details in the proof of Theorem 15.
2. Let R be an integral domain and let D be a nonempty subset of R that is closed under multiplication. Prove that the ring of fractions $D^{-1}R$ is isomorphic to a subring of the quotient field of R (hence is also an integral domain).
3. Let F be a field. Prove that F contains a unique smallest subfield F_0 and that F_0 is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$ for some prime p (F_0 is called the *prime subfield* of F). [See Exercise 26, Section 3.]
4. Prove that any subfield of $\mathbb{R}$ must contain $\mathbb{Q}$.

5. If F is a field, prove that the field of fractions of $F[[x]]$ (the ring of formal power series in the indeterminate x with coefficients in F) is the ring $F((x))$ of formal Laurent series (cf. Exercises 3 and 5 of Section 2). Show the field of fractions of the power series ring $\mathbb{Z}[[x]]$ is properly contained in the field of Laurent series $\mathbb{Q}((x))$. [Consider the series for e^x .]
6. Prove that the real numbers, $\mathbb{R}$, contain a subring A with $1 \in A$ and A maximal (under inclusion) with respect to the property that $\frac{1}{2} \notin A$. [Use Zorn's Lemma.] (Exercise 13 in Section 15.3 shows $\mathbb{R}$ is the quotient field of A , so $\mathbb{R}$ is the quotient field of a proper subring.)

7.6 THE CHINESE REMAINDER THEOREM

Throughout this section we shall assume unless otherwise stated that all rings are commutative with an identity $1 \neq 0$.

Given an arbitrary collection of rings (not necessarily satisfying the conventions above), their (*ring*) *direct product* is defined to be their direct product as (abelian) groups made into a ring by defining multiplication componentwise. In particular, if R_1 and R_2 are two rings, we shall denote by $R_1 \times R_2$ their direct product (as rings), that is, the set of ordered pairs (r_1, r_2) with $r_1 \in R_1$ and $r_2 \in R_2$ where addition and multiplication are performed componentwise:

$$(r_1, r_2) + (s_1, s_2) = (r_1 + s_1, r_2 + s_2) \quad \text{and} \quad (r_1, r_2)(s_1, s_2) = (r_1 s_1, r_2 s_2).$$

We note that a map φ from a ring R into a direct product ring is a homomorphism if and only if the induced maps into each of the components are homomorphisms.

There is a generalization to arbitrary rings of the notion in $\mathbb{Z}$ of two integers n and m being relatively prime (even to rings where the notion of greatest common divisor is not defined). In $\mathbb{Z}$ this is equivalent to being able to solve the equation $nx + my = 1$ in integers x and y (this fact was stated in Chapter 0 and will be proved in Chapter 8). This in turn is equivalent to $n\mathbb{Z} + m\mathbb{Z} = \mathbb{Z}$ as ideals (in general, $n\mathbb{Z} + m\mathbb{Z} = (m, n)\mathbb{Z}$). This motivates the following definition:

Definition. The ideals A and B of the ring R are said to be *comaximal* if $A + B = R$.

Recall that the *product*, AB , of the ideals A and B of R is the ideal consisting of all finite sums of elements of the form xy , $x \in A$ and $y \in B$ (cf. Exercise 34, Section 3). If $A = (a)$ and $B = (b)$, then $AB = (ab)$. More generally, the product of the ideals $A_1, A_2, \dots, A_k$ is the ideal of all finite sums of elements of the form $x_1 x_2 \cdots x_k$ such that $x_i \in A_i$ for all i . If $A_i = (a_i)$, then $A_1 \cdots A_k = (a_1 \cdots a_k)$.

Theorem 17. (Chinese Remainder Theorem) Let $A_1, A_2, \dots, A_k$ be ideals in R . The map

$$R \rightarrow R/A_1 \times R/A_2 \times \cdots \times R/A_k \quad \text{defined by} \quad r \mapsto (r + A_1, r + A_2, \dots, r + A_k)$$

is a ring homomorphism with kernel $A_1 \cap A_2 \cap \cdots \cap A_k$. If for each $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$ the ideals A_i and A_j are comaximal, then this map is surjective and $A_1 \cap A_2 \cap \cdots \cap A_k = A_1 A_2 \cdots A_k$, so

$$R/(A_1 A_2 \cdots A_k) = R/(A_1 \cap A_2 \cap \cdots \cap A_k) \cong R/A_1 \times R/A_2 \times \cdots \times R/A_k.$$

Proof: We first prove this for $k = 2$; the general case will follow by induction. Let $A = A_1$ and $B = A_2$. Consider the map $\varphi : R \rightarrow R/A \times R/B$ defined by $\varphi(r) = (r \bmod A, r \bmod B)$, where $\bmod A$ means the class in R/A containing r (that is, $r + A$). This map is a ring homomorphism because φ is just the natural projection of R into R/A and R/B for the two components. The kernel of φ consists of all the elements $r \in R$ that are in A and in B , i.e., $A \cap B$. To complete the proof in this case it remains to show that when A and B are comaximal, φ is surjective and $A \cap B = AB$. Since $A + B = R$, there are elements $x \in A$ and $y \in B$ such that $x + y = 1$. This equation shows that $\varphi(x) = (0, 1)$ and $\varphi(y) = (1, 0)$ since, for example, x is an element of A and $x = 1 - y \in 1 + B$. If now $(r_1 \bmod A, r_2 \bmod B)$ is an arbitrary element in $R/A \times R/B$, then the element $r_2x + r_1y$ maps to this element since

$$\begin{aligned}\varphi(r_2x + r_1y) &= \varphi(r_2)\varphi(x) + \varphi(r_1)\varphi(y) \\ &= (r_2 \bmod A, r_2 \bmod B)(0, 1) + (r_1 \bmod A, r_1 \bmod B)(1, 0) \\ &= (0, r_2 \bmod B) + (r_1 \bmod A, 0) \\ &= (r_1 \bmod A, r_2 \bmod B).\end{aligned}$$

This shows that φ is indeed surjective. Finally, the ideal AB is always contained in $A \cap B$. If A and B are comaximal and x and y are as above, then for any $c \in A \cap B$, $c = c1 = cx + cy \in AB$. This establishes the reverse inclusion $A \cap B \subseteq AB$ and completes the proof when $k = 2$.

The general case follows easily by induction from the case of two ideals using $A = A_1$ and $B = A_2 \cdots A_k$ once we show that A_1 and $A_2 \cdots A_k$ are comaximal. By hypothesis, for each $i \in \{2, 3, \dots, k\}$ there are elements $x_i \in A_1$ and $y_i \in A_i$ such that $x_i + y_i = 1$. Since $x_i + y_i \equiv y_i \bmod A_1$, it follows that $1 = (x_2 + y_2) \cdots (x_k + y_k)$ is an element in $A_1 + (A_2 \cdots A_k)$. This completes the proof.

This theorem obtained its name from the special case $\mathbb{Z}/mn\mathbb{Z} \cong (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ as rings when m and n are relatively prime integers. We proved this isomorphism just for the additive groups earlier. This isomorphism, phrased in number-theoretic terms, relates to simultaneously solving two congruences modulo relatively prime integers (and states that such congruences can always be solved, and uniquely). Such problems were considered by the ancient Chinese, hence the name. Some examples are provided in the exercises.

Since the isomorphism in the Chinese Remainder Theorem is an isomorphism of rings, in particular the groups of units on both sides must be isomorphic. It is easy to see that the units in any direct product of rings are the elements that have units in each of the coordinates. In the case of $\mathbb{Z}/mn\mathbb{Z}$ the Chinese Remainder Theorem gives the following isomorphism on the groups of units:

$$(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

More generally we have the following result.